



YFG FCL LIGHT TEST.

Operator / Maintenance Manual

Manufactured by:
Paveway Automation Pvt. Ltd.
Sanghvi Steel Complex,
128/2, TELCO Road,
Chinchwad,
Pune - 411019



YFG FCL LIGHT TEST.

Marelli Motherson Automotive Lighting India Pvt Ltd.
Gat No-165/1,165/2 &165/3 Village Ambethan,
Dwarka School, Taluka Khed, Chakan, Pune-410501.



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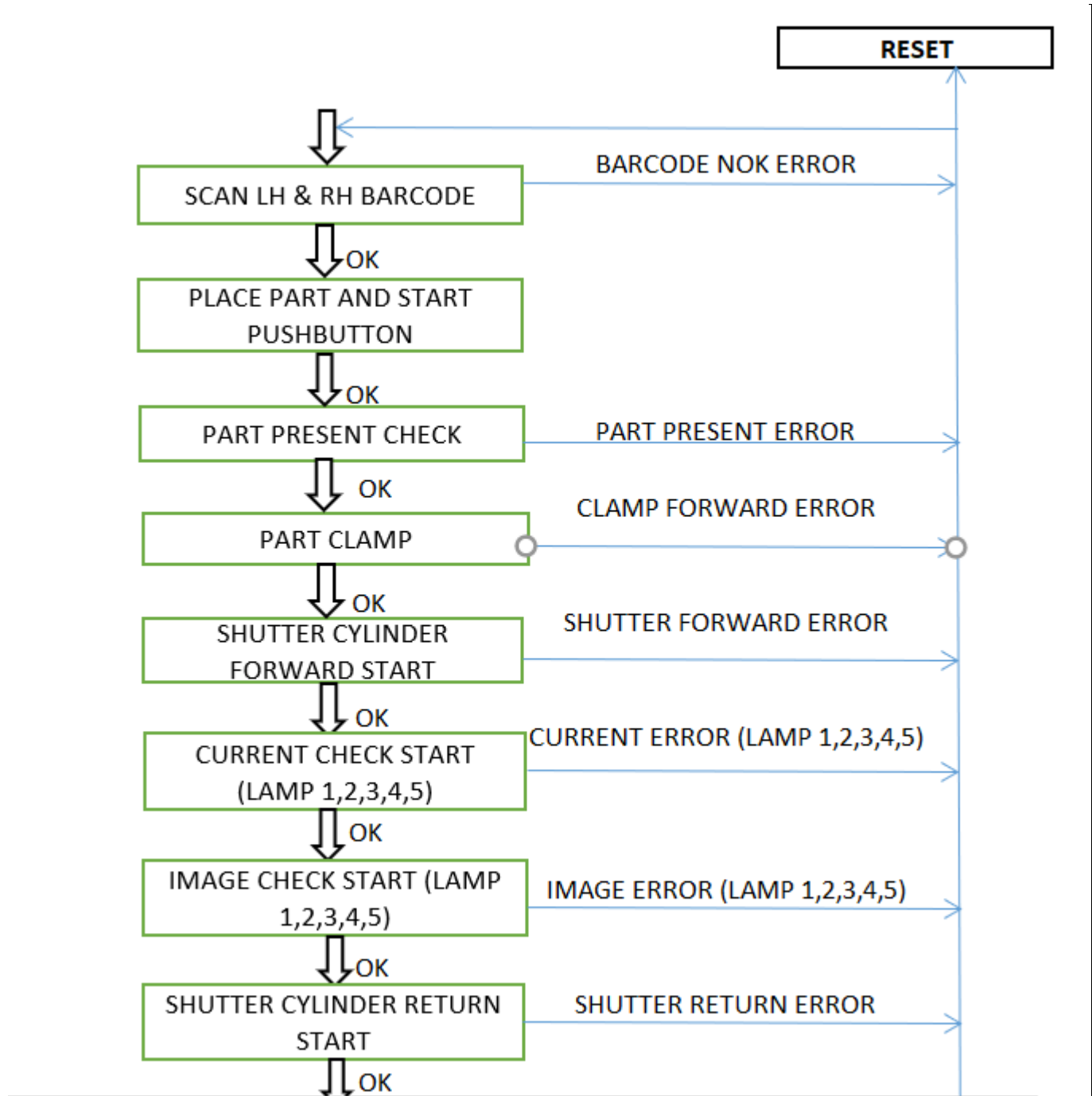
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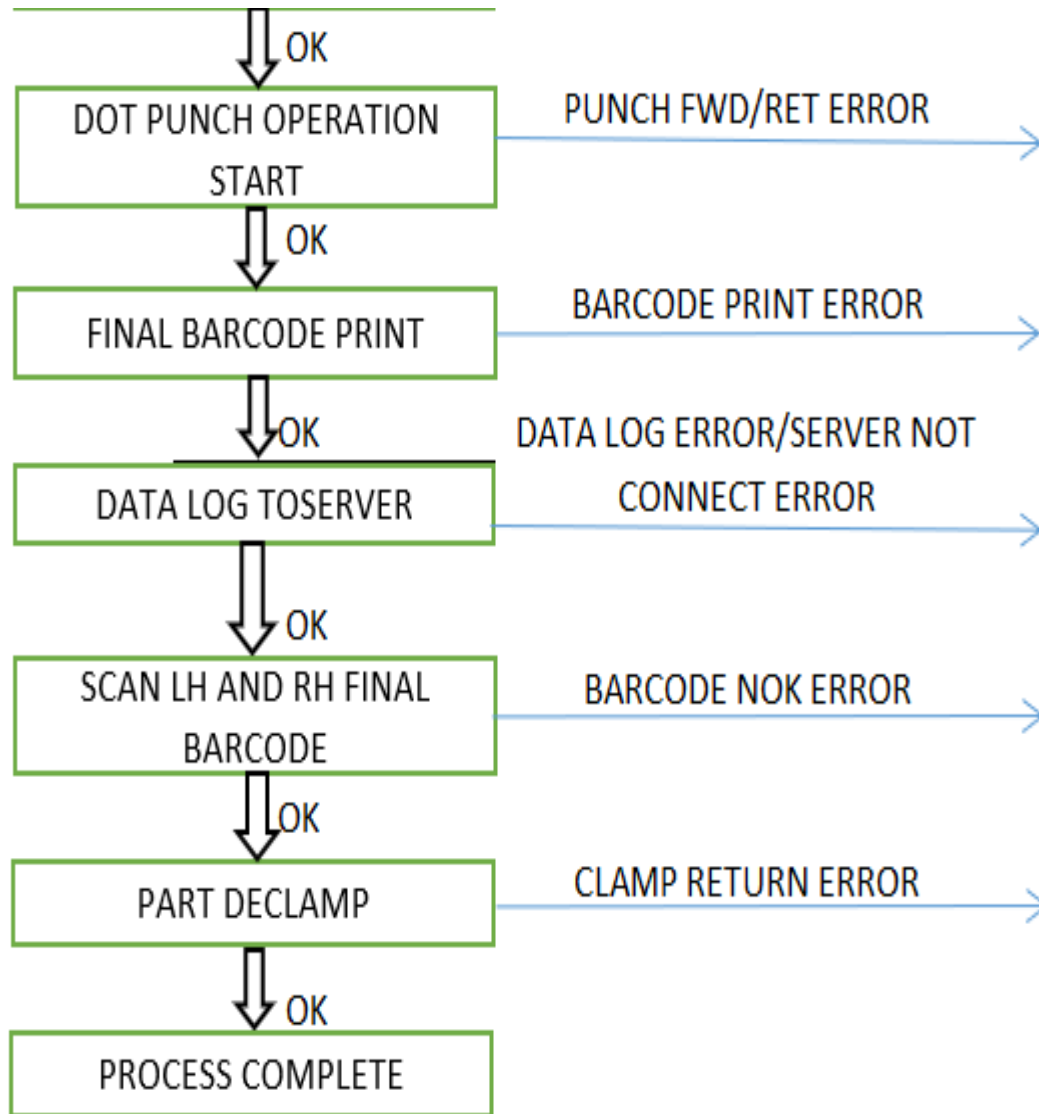


SYSTEM INFORMATION:

Name :	YFG FCL LIGHT TEST.
Customer :	Marelli Motherson Automotive Lighting India Pvt Ltd. Gat No-165/1,165/2 &165/3 Village Ambethan, Dwarka School, Taluka Khed, Chakan, Pune-410501.
Manufacturing Date :	June, 2022.
Supply Voltage :	230 VAC, 50/60 Hz
Power consumption:	0.674KW
Manufacturer :	Paveway Automation Pvt. Ltd. Sanghvi Steel Complex, 128/2, Telco Road, Chinchwad, Pune- 411019. tel: 020-27450840
Email ID :	applications@pavewayautomation.com

PROCESS SEQUENCE:







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PRECAUTION STEPS:

How to start

Follow steps.

- 1) Power on main control panel.
- 2) Power on UPS.
- 3) Power on CPU.
- 4) Wait for windows booting process.
- 5) After windows boot yfg light test application will open automatically. Put user name and password to open light test application.
- 6) Now application is ready to use.

How to close

Follow steps.

- 1) After completing production close application properly.
- 2) Then shut down CPU.
- 3) After shut down CPU power off UPS.
- 4) Then power off control panel.

Note: Take precaution while power on and off CPU. Direct power off without shut down will causes system crash.

YFG FCL LIGHT TEST.

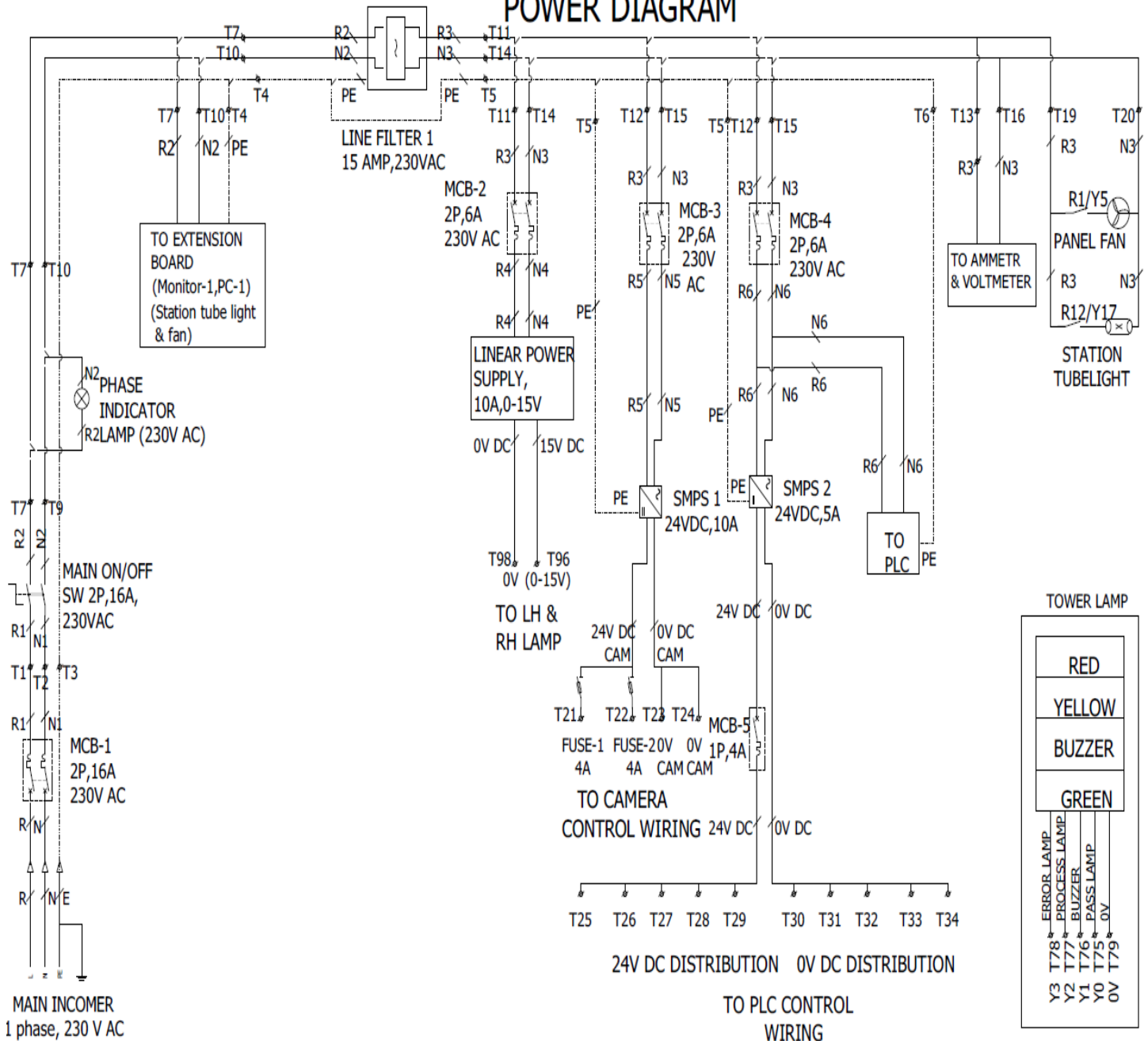
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**WIRING DIAGRAMS:****SYMBOL DESCRIPTION****SYMBOL DESCRIPTION**

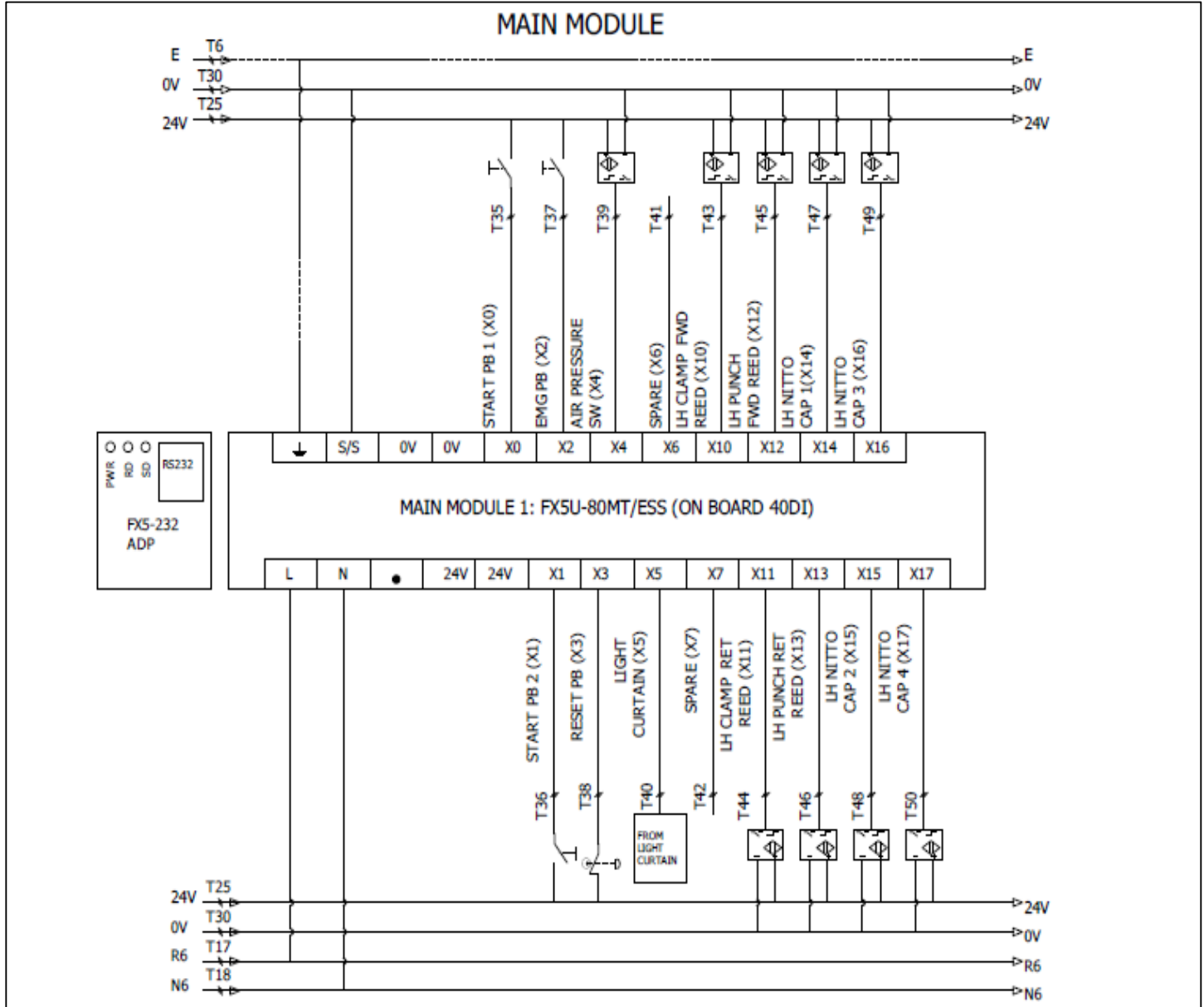
Symbol	Symbol description	Symbol	Symbol description
	Earthing		Proximity sensor
	2 Pole MCB		Buzzer
	2 Pole on/off switch		Double acting solenoid valve
	Lamp		Double acting solenoid valve (pneumatic)
	Tubelight		Single acting solenoid valve (pneumatic)
	Fan		single acting solenoid valve
	SMPS		
	1 pole MCB		
	Push button		
	Relay coil		
	Emergency stop button		
	Airoff		

POWER DIAGRAM

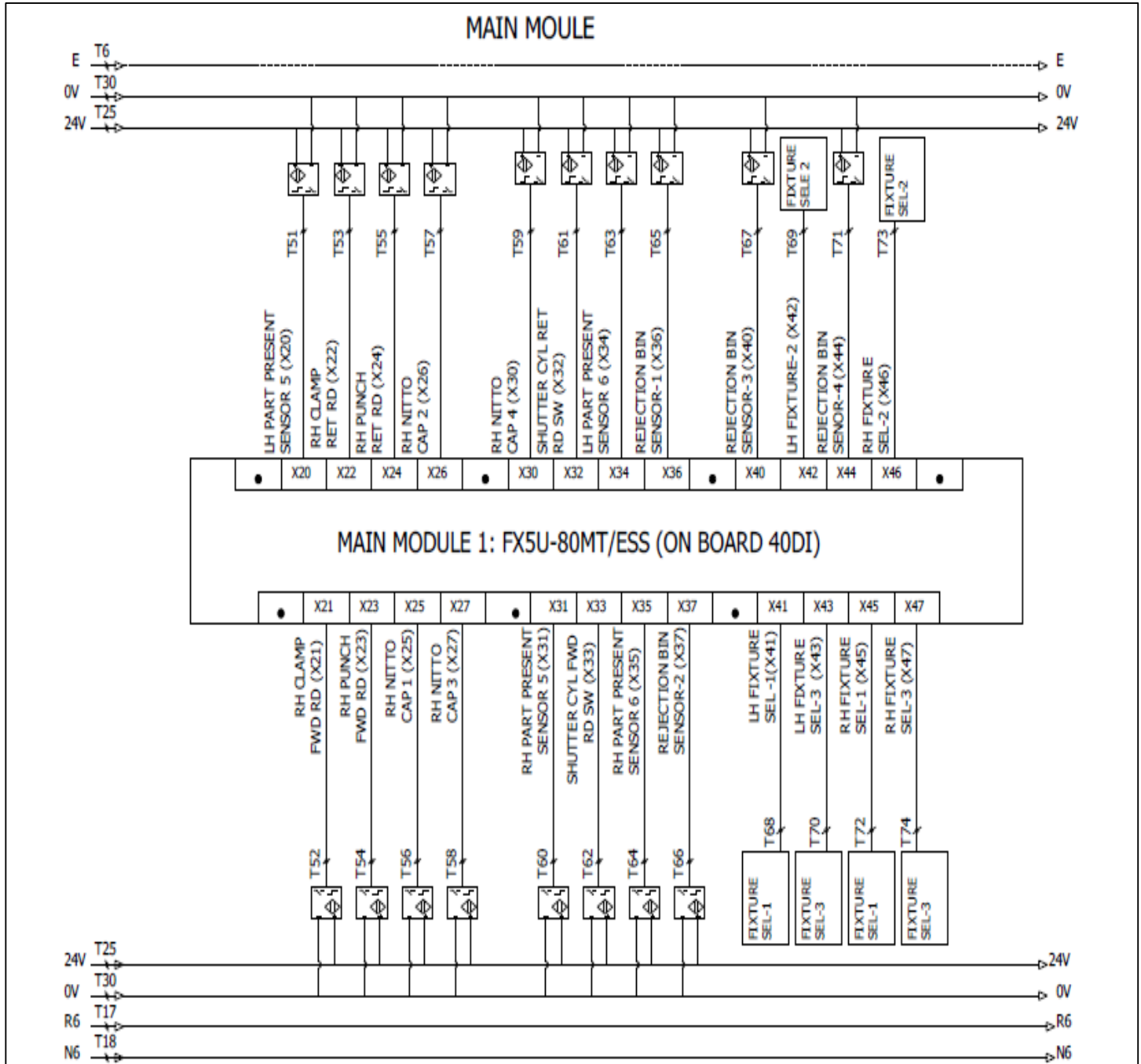
POWER DIAGRAM



MAINMODULE 1-FX5U-80MT/ESS (ON BOARD 40DI)

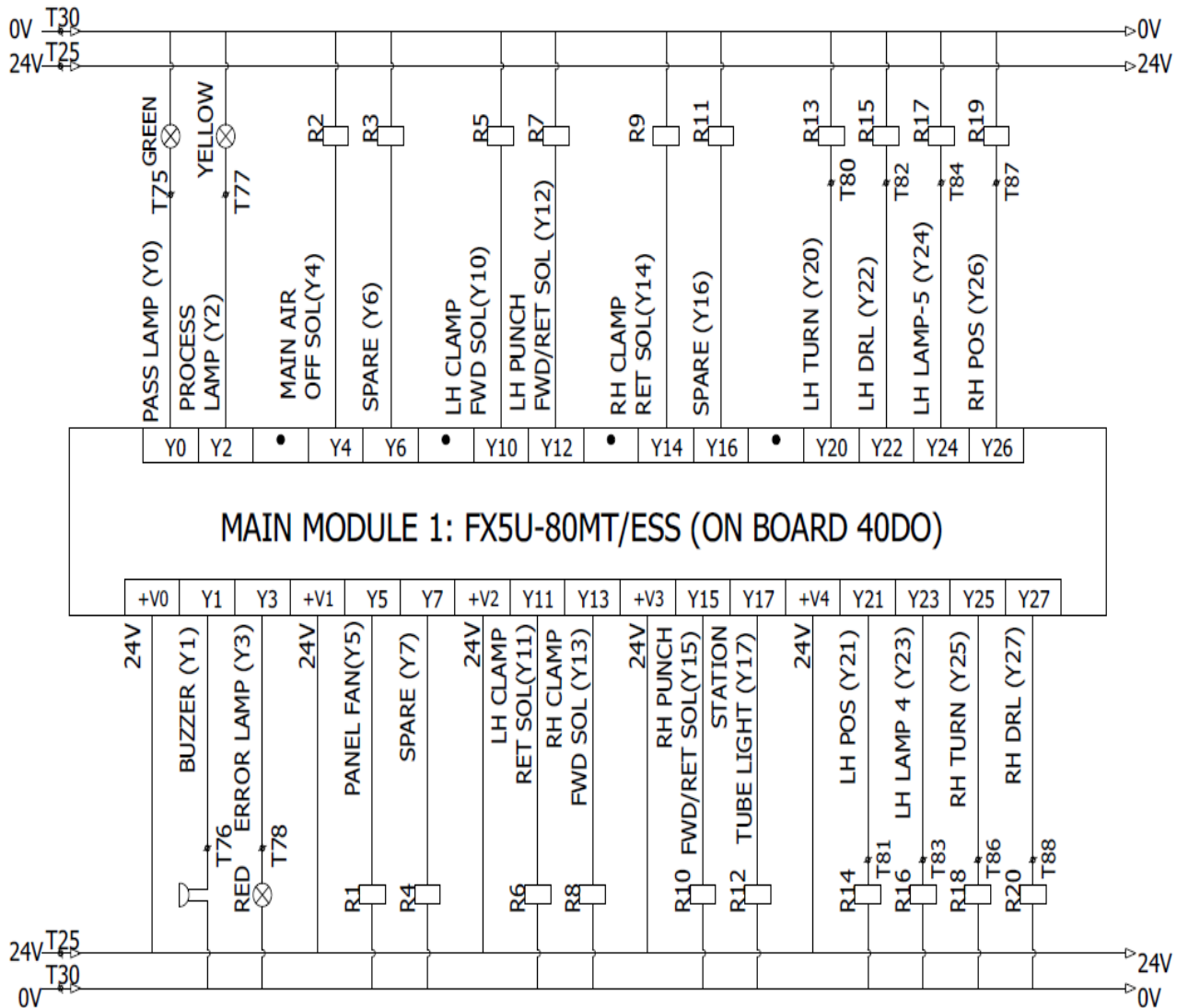


MAIN MODULE1: FX5U80MT/ESS (ON BOARD 40DI)



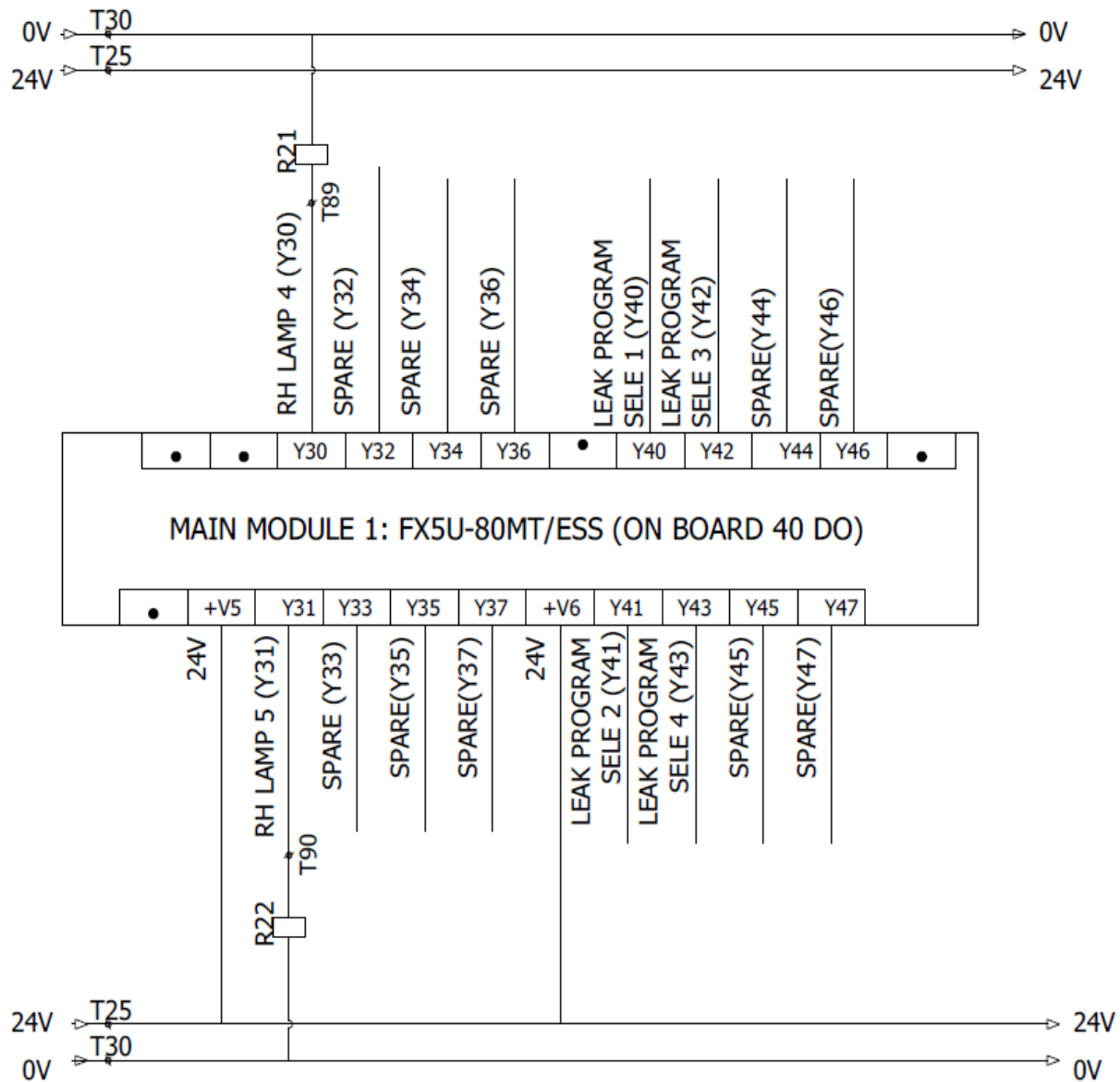
MAIN MODULE1: FX5U-80MT/ESS (ON BOARD 40DO)

MAIN MODULE



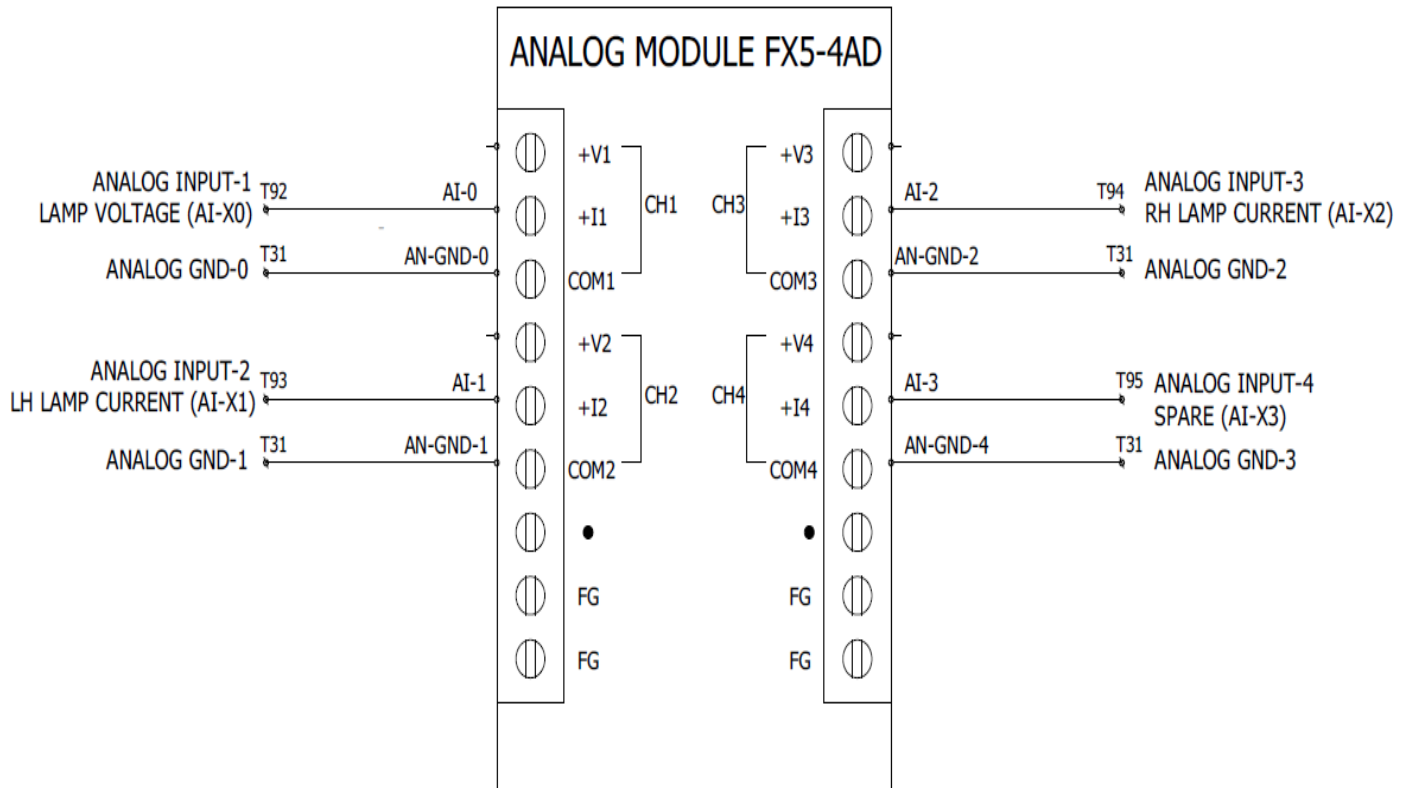
MAIN MODULE1: FX5U-80MT/ESS (ON BOARD 40DO)

MAIN MODULE



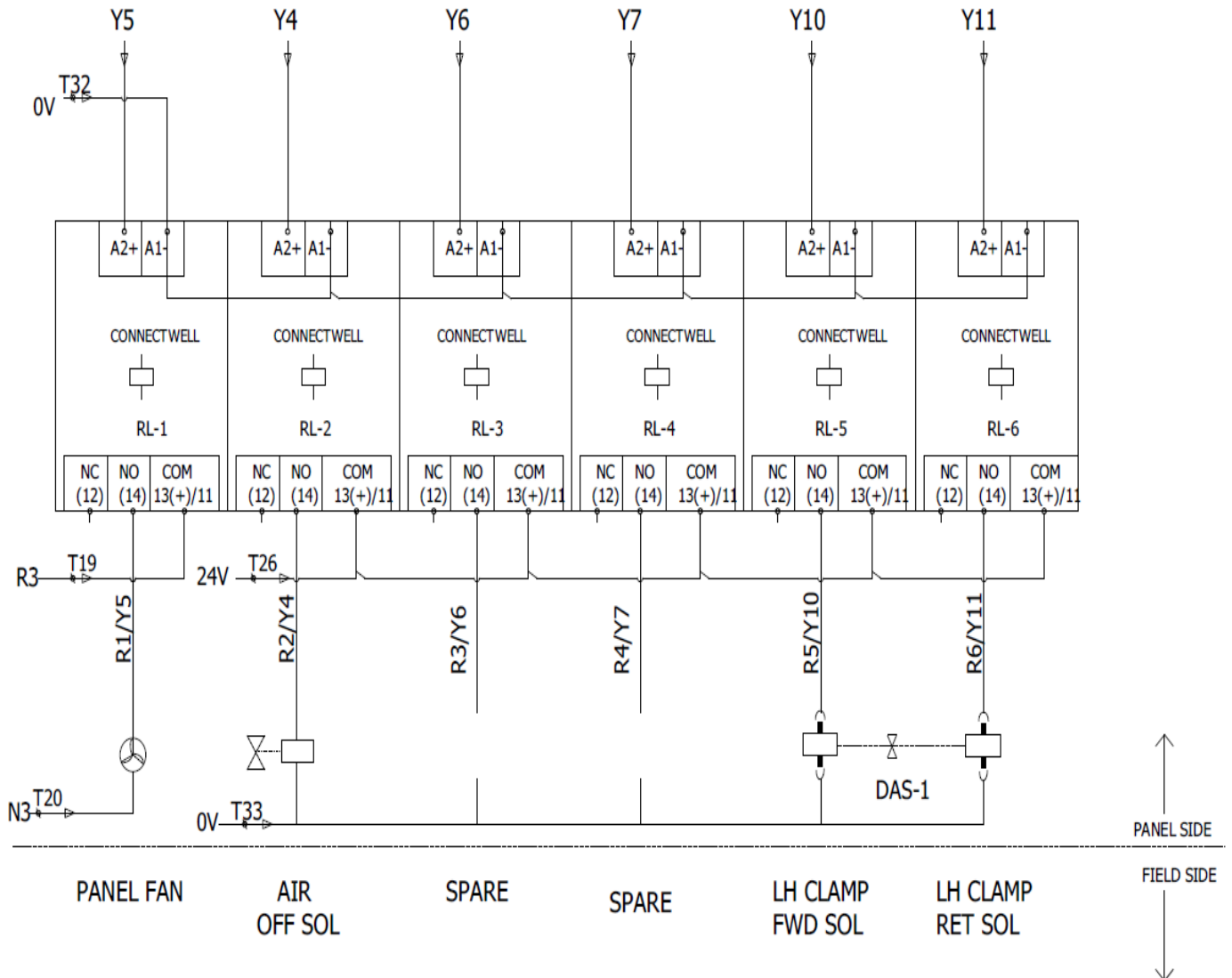
ANALOG MODULE FX5-4AD

ANALOG MODULE FX5-4AD INPUT



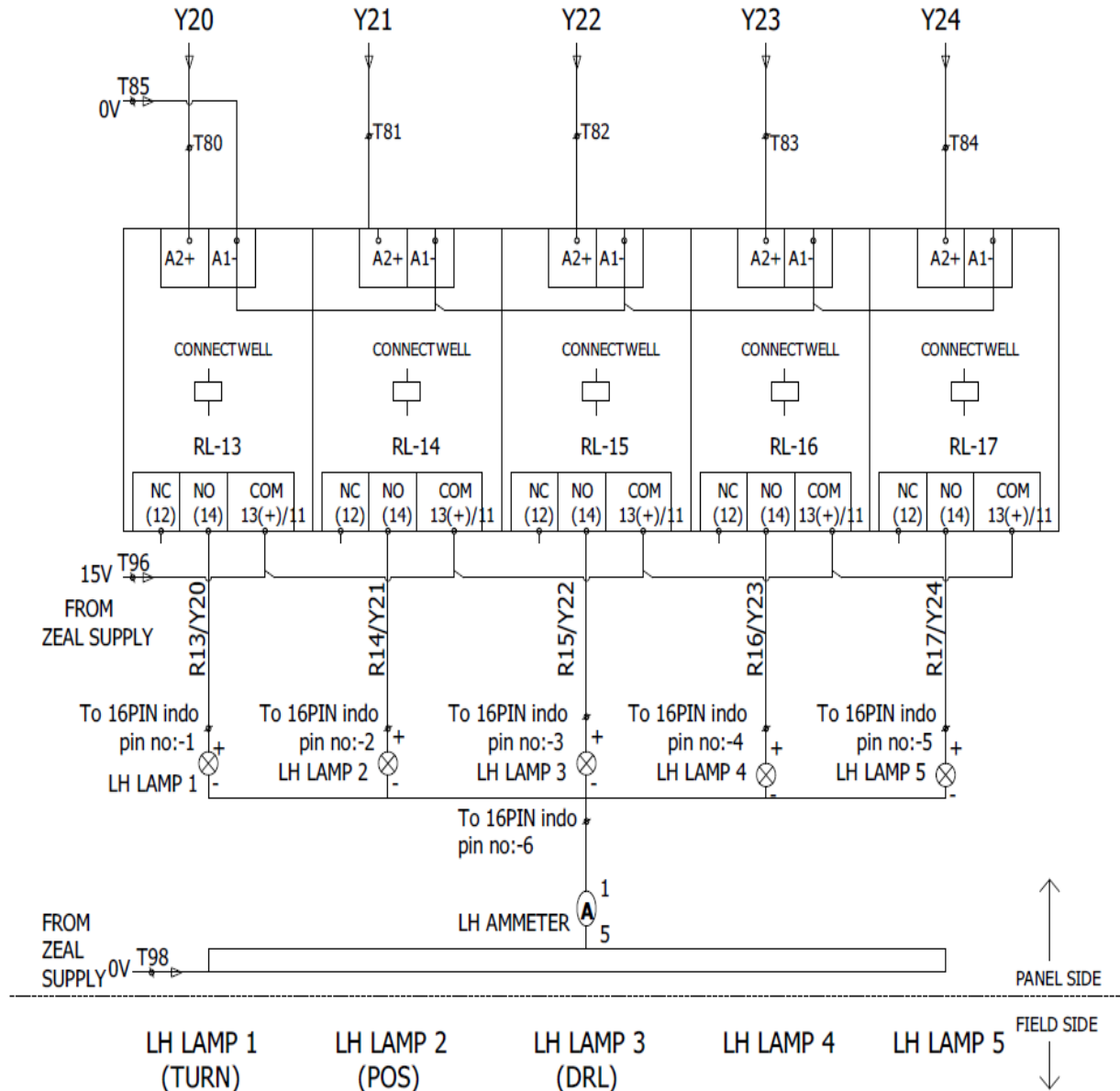
RELAY DIAGRAM-1

RELAY DIAGRAM-1



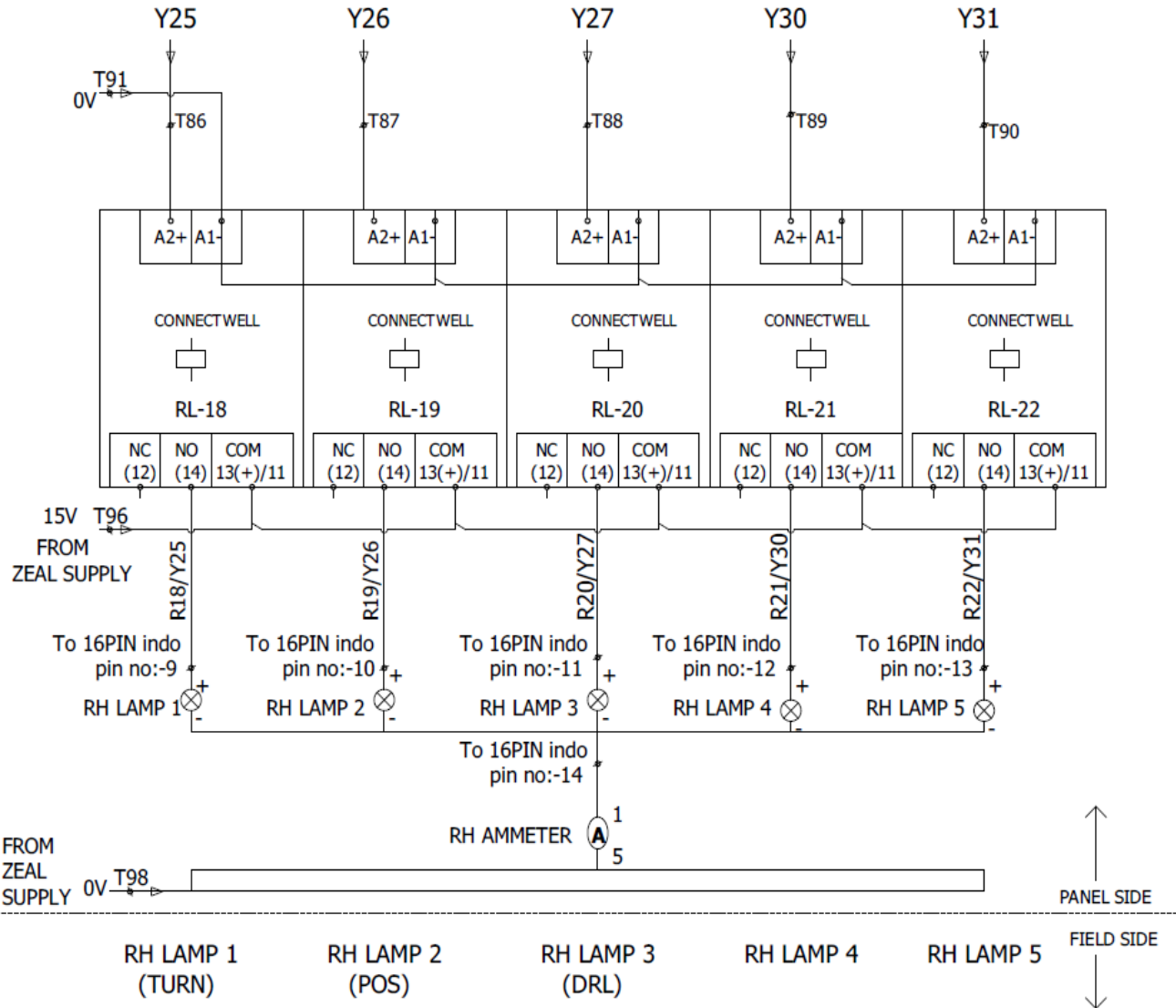
RELAY DIAGRAM-2

LH LAMP RELAY DIAGRAM-3



RH LAMP RELAY DIAGRAM-4

RH LAMP RELAY DIAGRAM-4



PANEL CONNECTOR DETAILS-1

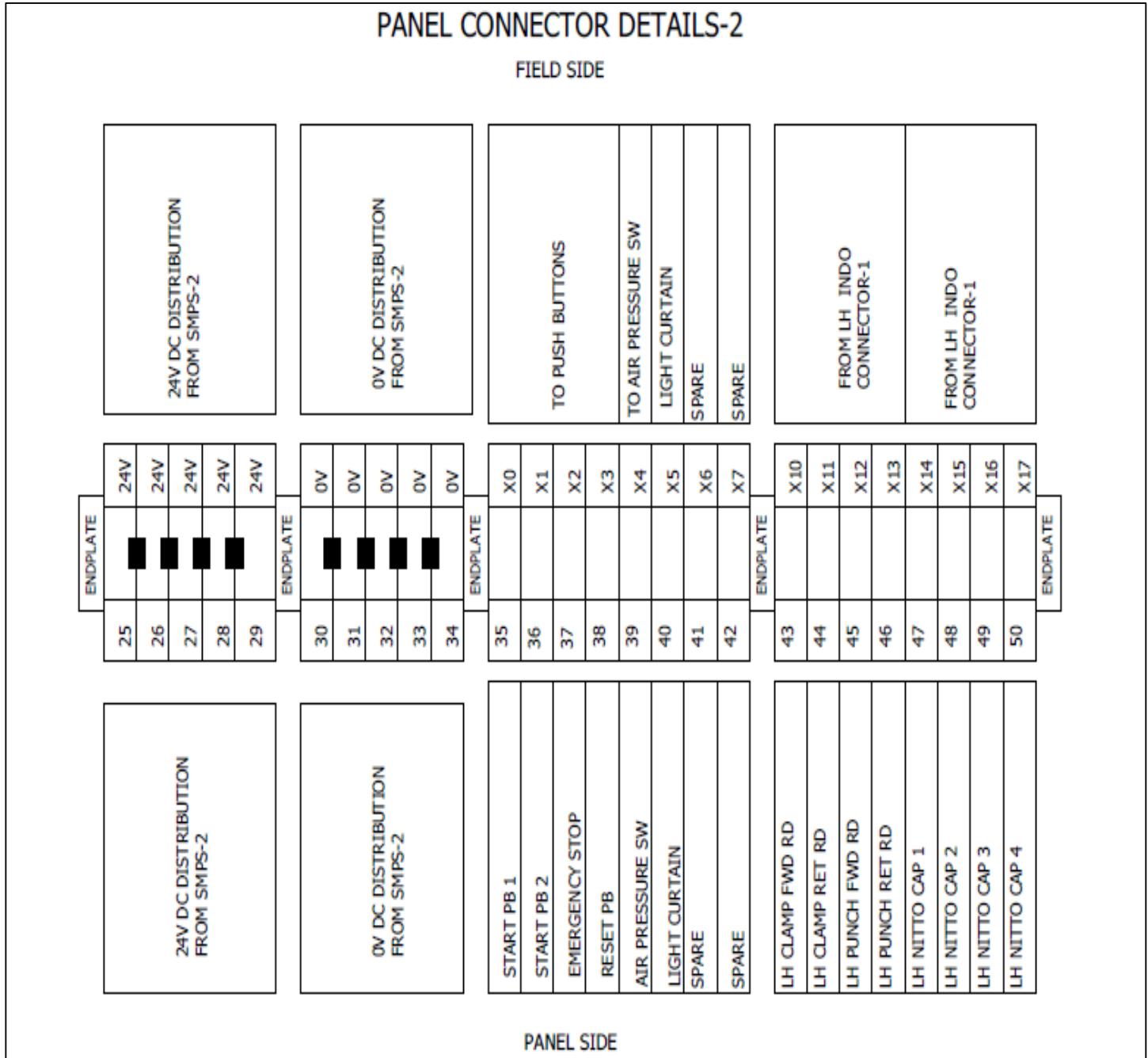
PANEL CONNECTOR DETAILS-1

FIELD SIDE

ENDSTOPPER		1		R1	LINE DISTRIBUTION
		2		N1	NEUTRAL DISTRIBUTION
ENDPLATE		3		E	EARTH DISTRIBUTION
		4		E	
		5		E	
		6		E	
ENDPLATE		7		R2	LINE DISTRIBUTION
		8		R2	
ENDPLATE		9		N2	NEUTRAL DISTRIBUTION
		10		N2	
ENDPLATE		11		R3	LINE DISTRIBUTION
		12		R3	
		13		R3	
ENDPLATE		14		N3	NEUTRAL DISTRIBUTION
		15		N3	
		16		N3	
ENDPLATE		17		R5	LINE DISTRIBUTION
		18		N5	NEUTRAL DISTRIBUTION
ENDPLATE		19		R1/Y5	TO PANEL FAN
		20		N3	
ENDPLATE		21		FUSE 1	24V DC DISTRIBUTION FROM FUSE TERMINAL TO CAMERA CONTROL
		22		FUSE 2	
ENDPLATE		23		0V CAM	SMPS-1 0V DC DISTRIBUTION FOR CAMERA CONTROLLER
		24		0V CAM	
ENDPLATE					

PANEL SIDE

PANEL CONNECTOR DETAILS-2



PANEL CONNECTOR DETAILS-3

PANEL CONNECTOR DETAILS-3

FIELD SIDE

ENDPLATE	51	X20	FROM LH INDO CONNECTOR-1
	52	X21	
	53	X22	
	54	X23	
	55	X24	
	56	X25	
	57	X26	
	58	X27	
ENDPLATE	59	X30	FROM RH INDO CONNECTOR-2
	60	X31	
	61	X32	
	62	X33	
	63	X34	
	64	X35	
	65	X36	
	66	X37	
ENDPLATE	67	X40	FROM LH INDO CONNECTOR-1
	68	X41	
	69	X42	
	70	X43	
	71	X44	
	72	X45	
	73	X46	
	74	X47	
ENDPLATE	75	Y0	TO TOWER LAMP
	76	Y1	
	77	Y2	
	78	Y3	
	79	0V	

PANEL SIDE

PANEL CONNECTOR DETAILS-4

PANEL CONNECTOR DETAILS-4

FIELD SIDE

ENDPLATE	80	Y20	TO 16 PIN INDO P.NO-1 (R13/Y20)
	81	Y21	TO 16 PIN INDO P.NO-2 (R14/Y21)
	82	Y22	TO 16PIN INDO P.NO-3 (R15/Y22)
	83	Y23	TO 16 PIN INDO P.NO-4 (R16/Y23)
	84	Y24	TO 16 PIN INDO P.NO-5 (R17/Y24)
	85	0V	0V DC DISTRIBUTIONFROM SMPS -2
ENDPLATE	86	Y25	TO 16 PIN INDO P.NO-9 (R18/Y25)
	87	Y26	TO 16 PIN INDO P.NO-10(R19/Y26)
	88	Y27	TO 16 PIN INDO P.NO-11(R20/Y27)
	89	Y30	TO 16 PIN INDO P.NO-12(R21/Y30)
	90	Y31	TO 16 PIN INDO P.NO-13(R22/Y31)
	91	0V	0V DC DISTRIBUTION FROM SMPS-2
ENDPLATE	92	AI-0	TO VOLTMETER
	93	AI-1	TO LH AMMETER
	94	AI-2	TO RH AMMETER
	95	AI-3	SPARE
ENDPLATE	96	0-15V	(0-15V) FROM ZEAL SUPPLY
	97	0-15V	
	98	0V	0V FROM ZEAL SUPPLY
	99	0V	
ENDSTOPPER			

LH LAMP-1 (TURN)	
LH LAMP-2 (POS)	
LH LAMP-3 (DRL)	
LH LAMP-4	
LH LAMP-5	
0V DC DISTR. FROM SMPS-2	

RH LAMP-1 (TURN)	
RH LAMP-2 (POS)	
RH LAMP-3 (DRL)	
RH LAMP-4	
RH LAMP-5	
0V DC DISTR. FROM SMPS-2	

LAMP VOLTAGE	
LH LAMP CURRENT	
RH LAMP CURRENT	
SPARE	

(0-15V) FROM ZEAL SUPPLY	
0V FROM ZEAL SUPPLY	

PANEL SIDE



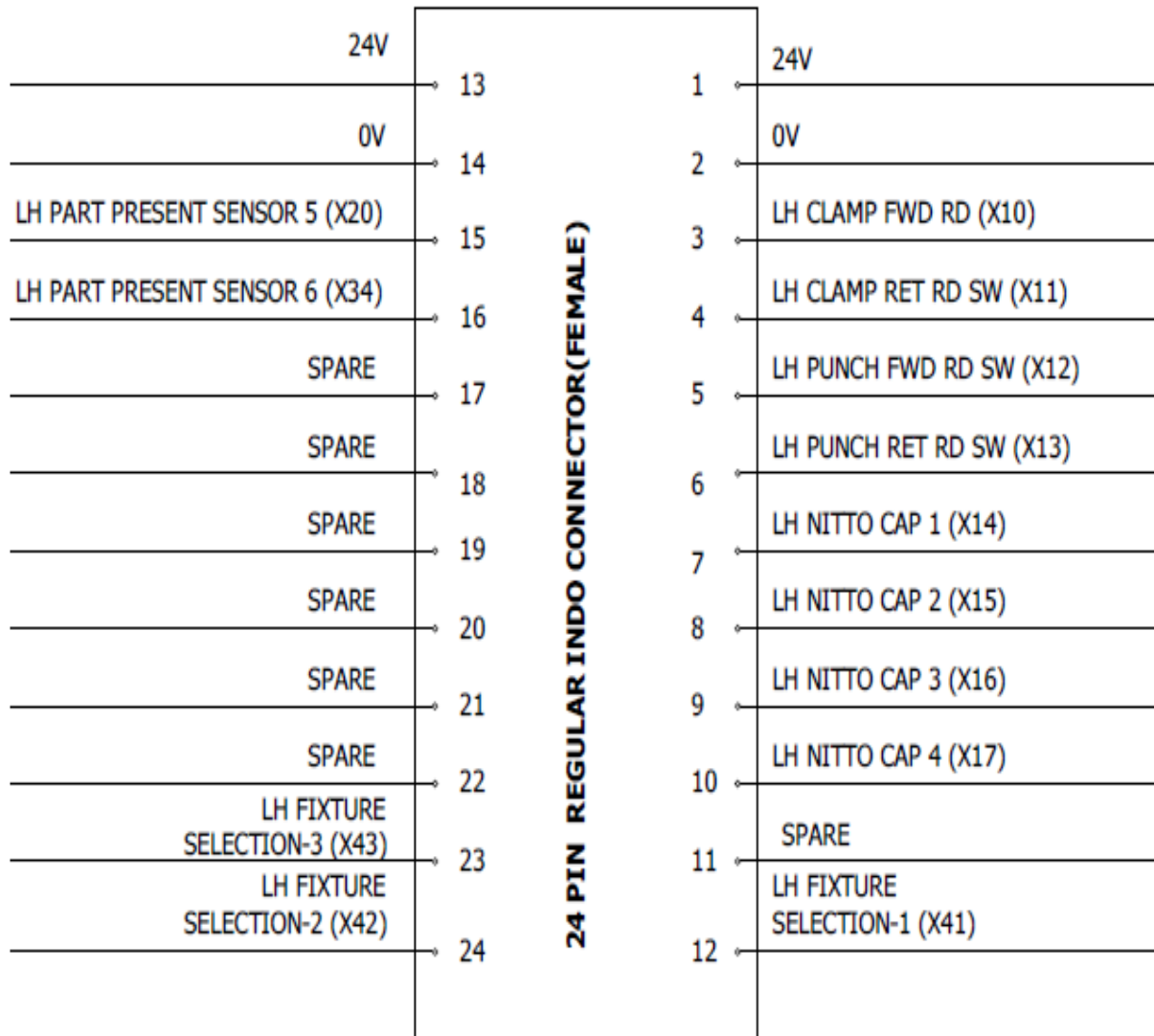
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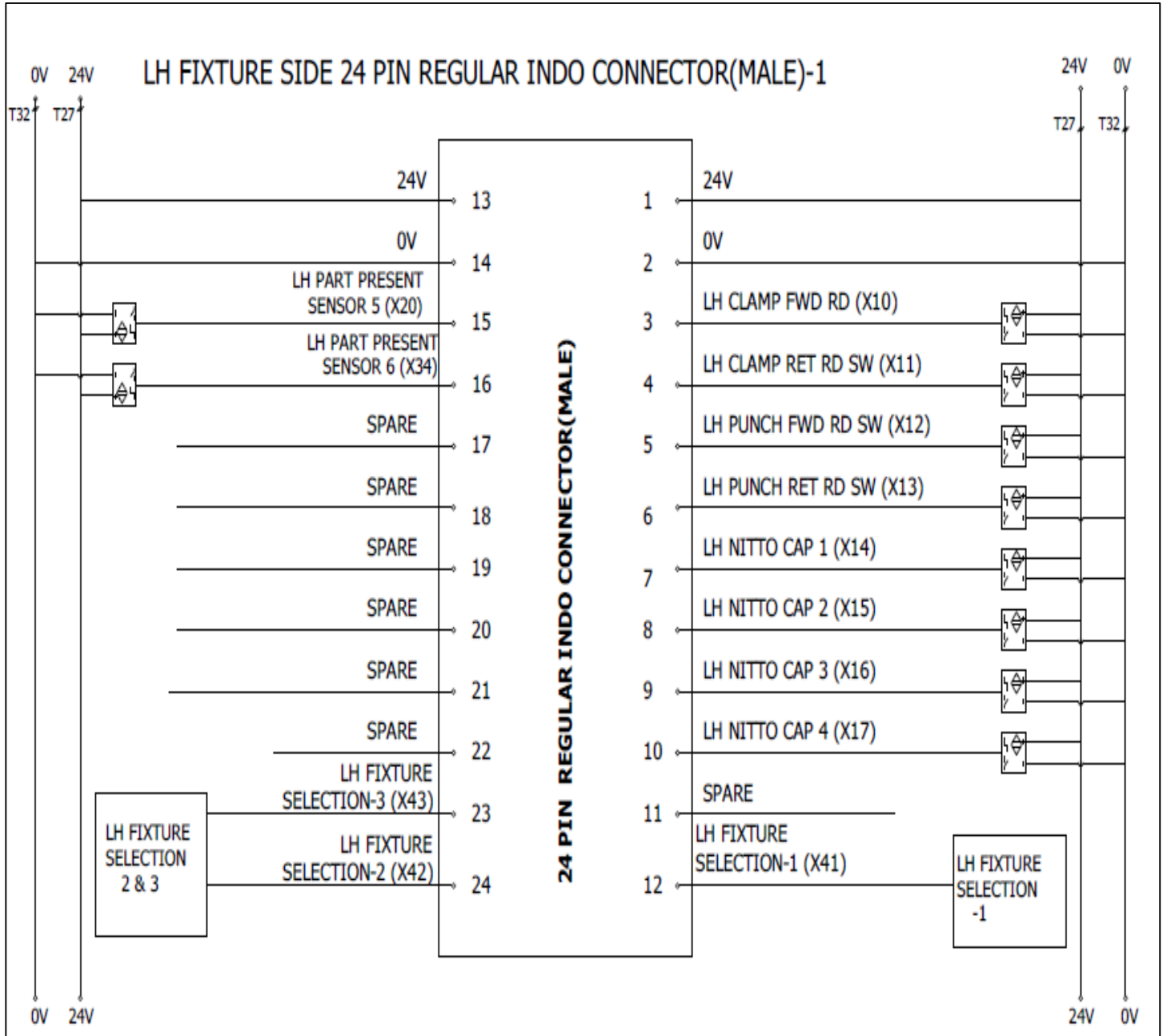


LH MACHINE SIDE 24 PIN INDO CONNECTOR (FEMALE)-1

LH MACHINE SIDE 24 PIN REGULAR INDO CONNECTOR(FEMALE)-1



LH FIXTURE SIDE 24 PIN INDO CONNECTOR (MALE)-1





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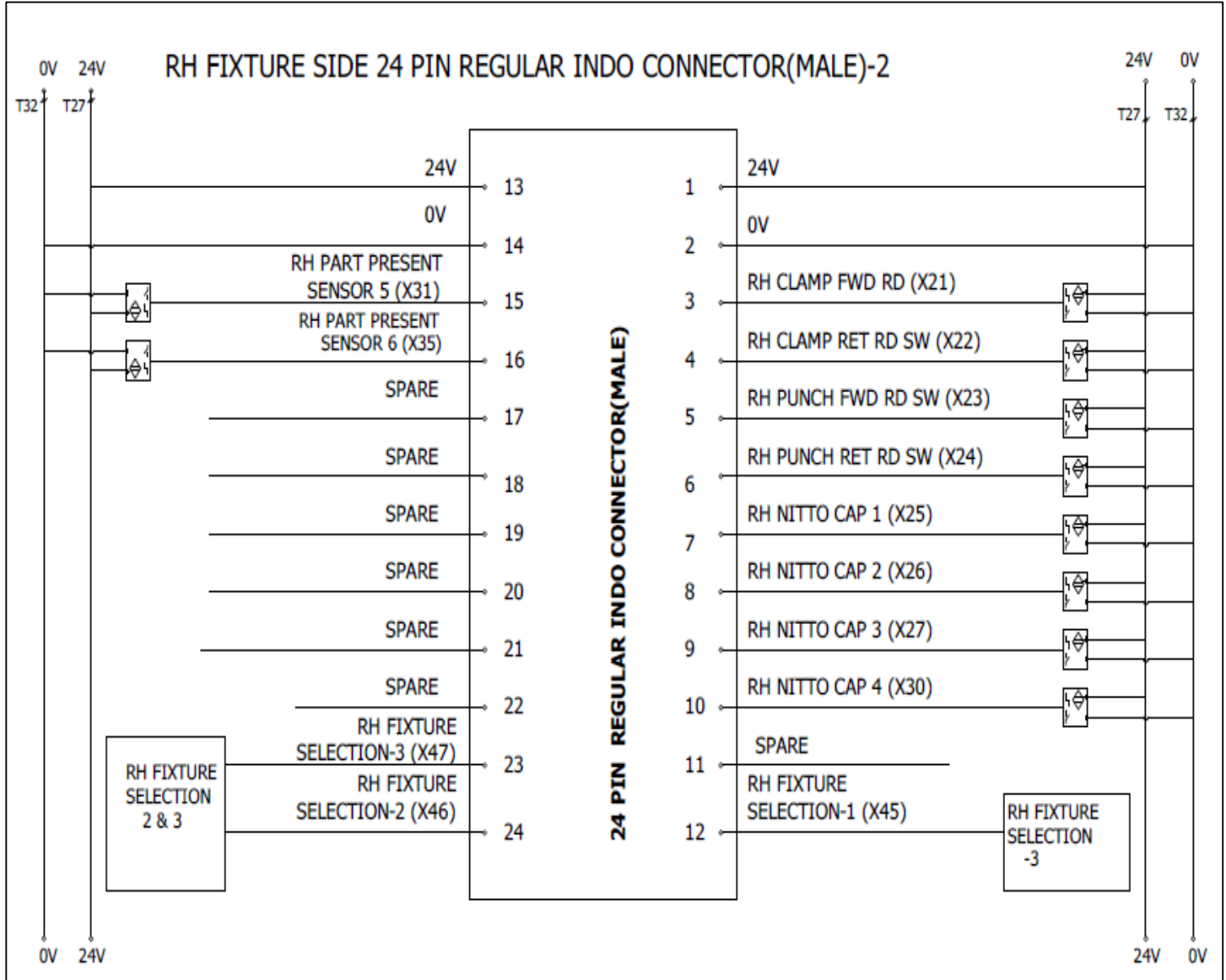
RH MACHINE SIDE 24 PIN INDO CONNECTOR (FEMALE)-2

RH MACHINE SIDE 24 PIN REGULAR INDO CONNECTOR(FEMALE)-2

24V	13	1	24V
0V	14	2	0V
RH PART PRESENT SENSOR 5 (X31)	15	3	RH CLAMP FWD RD (X21)
RH PART PRESENT SENSOR 6 (X35)	16	4	RH CLAMP RET RD SW (X22)
SPARE	17	5	RH PUNCH FWD RD SW (X23)
SPARE	18	6	RH PUNCH RET RD SW (X24)
SPARE	19	7	RH NITTO CAP 1 (X25)
SPARE	20	8	RH NITTO CAP 2 (X26)
SPARE	21	9	RH NITTO CAP 3 (X27)
SPARE	22	10	RH NITTO CAP 4 (X30)
RH FIXTURE SELECTION-4 (X46)	23	11	SPARE
RH FIXTURE SELECTION-4 (X46)	24	12	RH FIXTURE SELECTION-1 (X45)

24 PIN REGULAR INDO CONNECTOR(FEMALE)

RH FIXTURE SIDE 24 PIN INDO CONNECTOR (MALE)-2





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RH MACHINE SIDE 16 PIN INDO CONNECTOR (FEMALE)-3

MACHINE SIDE 16 PIN INDO CONNECTOR(FEMALE)-3

RH LAMP-1 (TURN)(R18/Y25)	9	16 PIN REGULAR INDO CONNECTOR(FEMALE)	1	LH LAMP-1 (TURN)(R13/Y20)
RH LAMP-2 (POS)(R19/Y26)	10		2	LH LAMP-2 (POS)(R14/Y21)
RH LAMP-3 (DRL)(R20/Y27)	11		3	LH LAMP-3 (DRL)(R15/Y22)
RH LAMP-4(R21/Y30)	12		4	LH LAMP-4(R16/Y23)
RH LAMP-5(R22/Y31)	13		5	LH LAMP-5(R17/Y24)
RH LAMP-	14		6	LH LAMP-
SPARE	15		7	SPARE
SPARE	16		8	SPARE



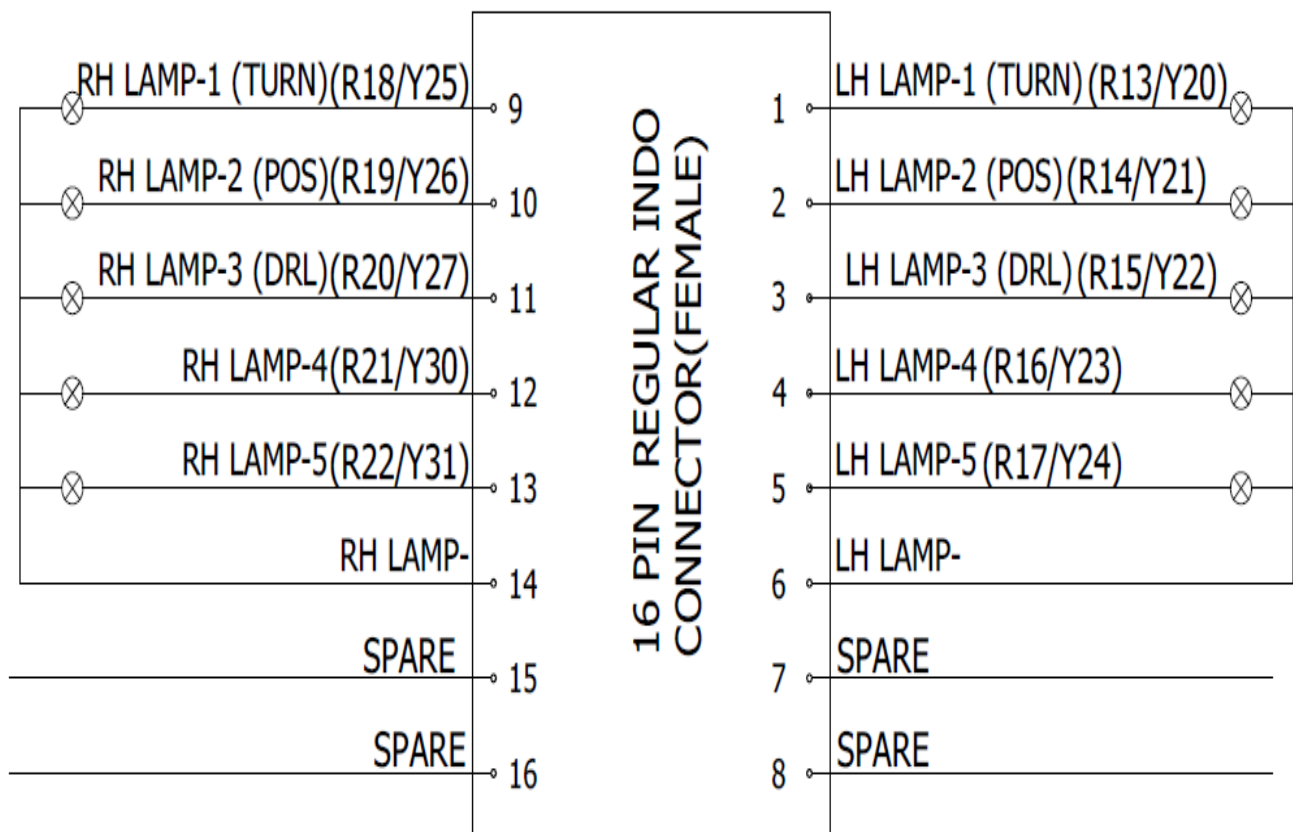
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FIXTURE SIDE 16 PIN INDO CONNECTOR (MALE)

FIXTURE SIDE 16 PIN INDO CONNECTOR(MALE)

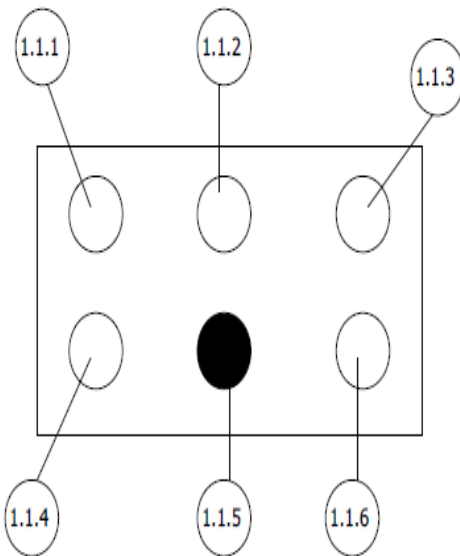


CONNECTOR DETAILS (SUMITOMO CONNECTOR (PART NO6188-0175))

CONNECTOR DETAILS

SUMITOMO CONNECTOR (PART NUMBER 6188-0175)

VIEW FROM WIRE
 INSERTION DIRECTION



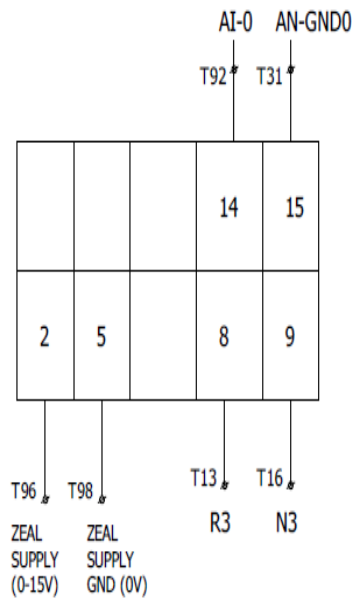
SUMITOMO CONNECTOR (PART NUMBER 6188-0175)

TURMINAL NUMBER	FUNCTION	COLOUR
1.1.1	TURN	YELLOW
1.1.2	POS	RED
1.1.3	DRL	BLUE
1.1.4	TURN GND	GREY
1.1.5	DUMMY	-----
1.1.6	DRL/POS GND	GREY

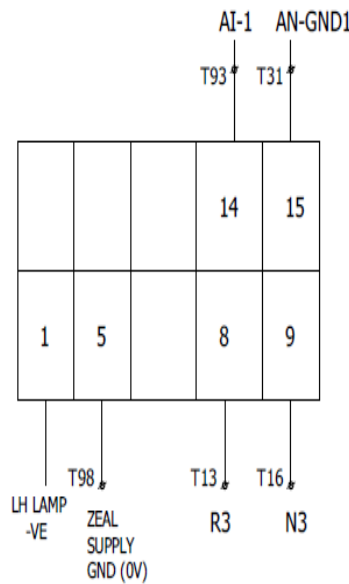
AMMETER AND VOLTMETER CONNECTIONS

AMMETER AND VOLTMETER CONNECTIONS

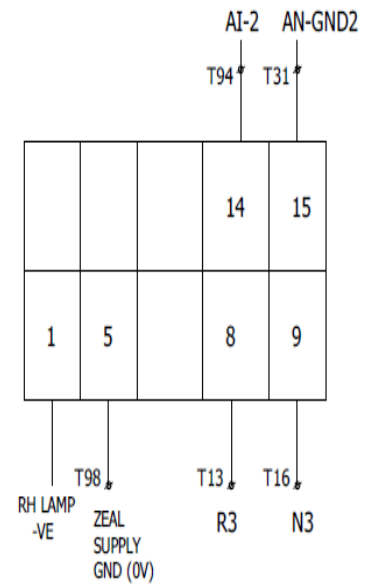
VOLTMETER (MT4W-DV)



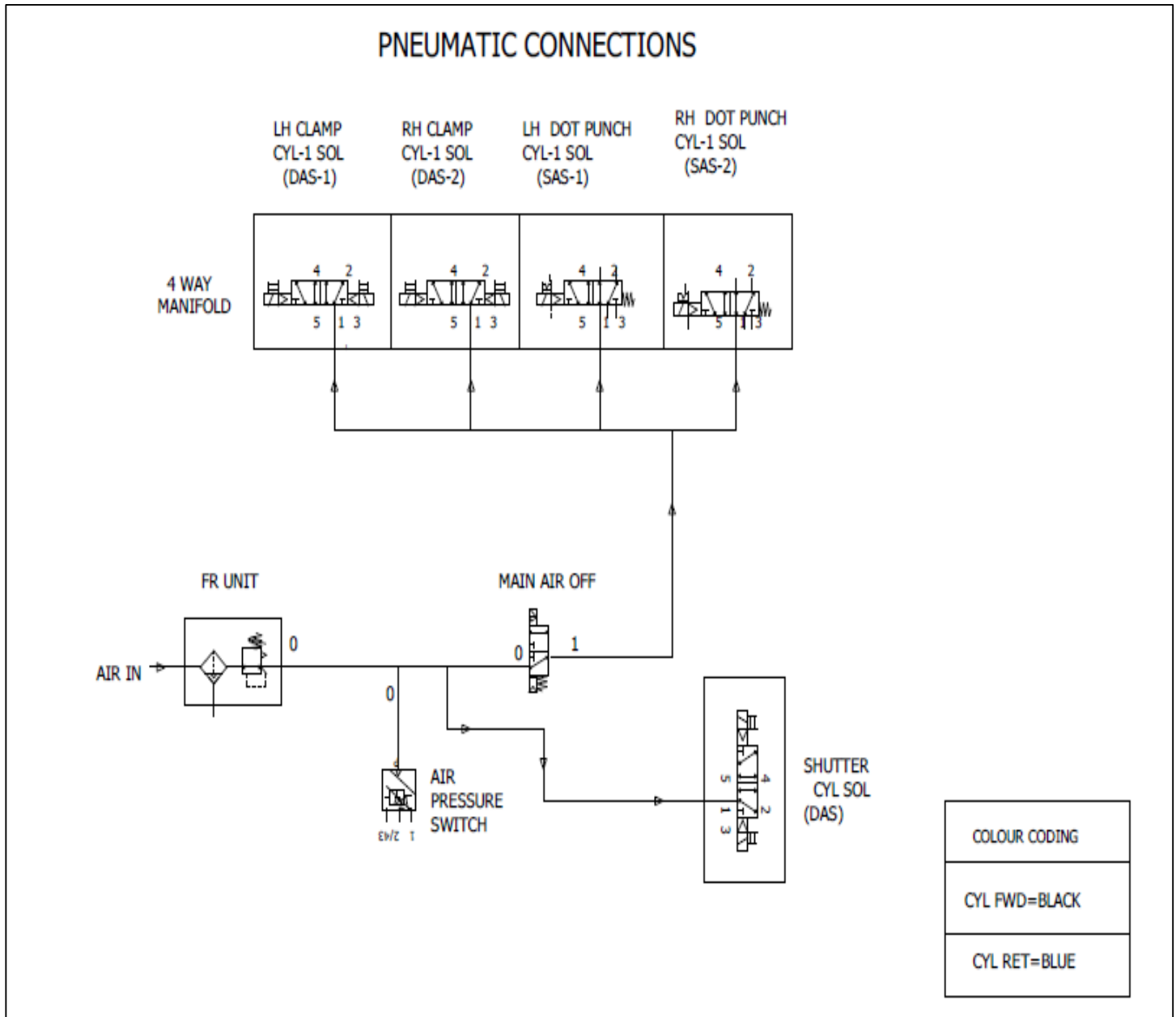
LH AMMETER (MT4W-DA)



RH AMMETER (MT4W-DA)



PNEUMATIC CONNECTIONS





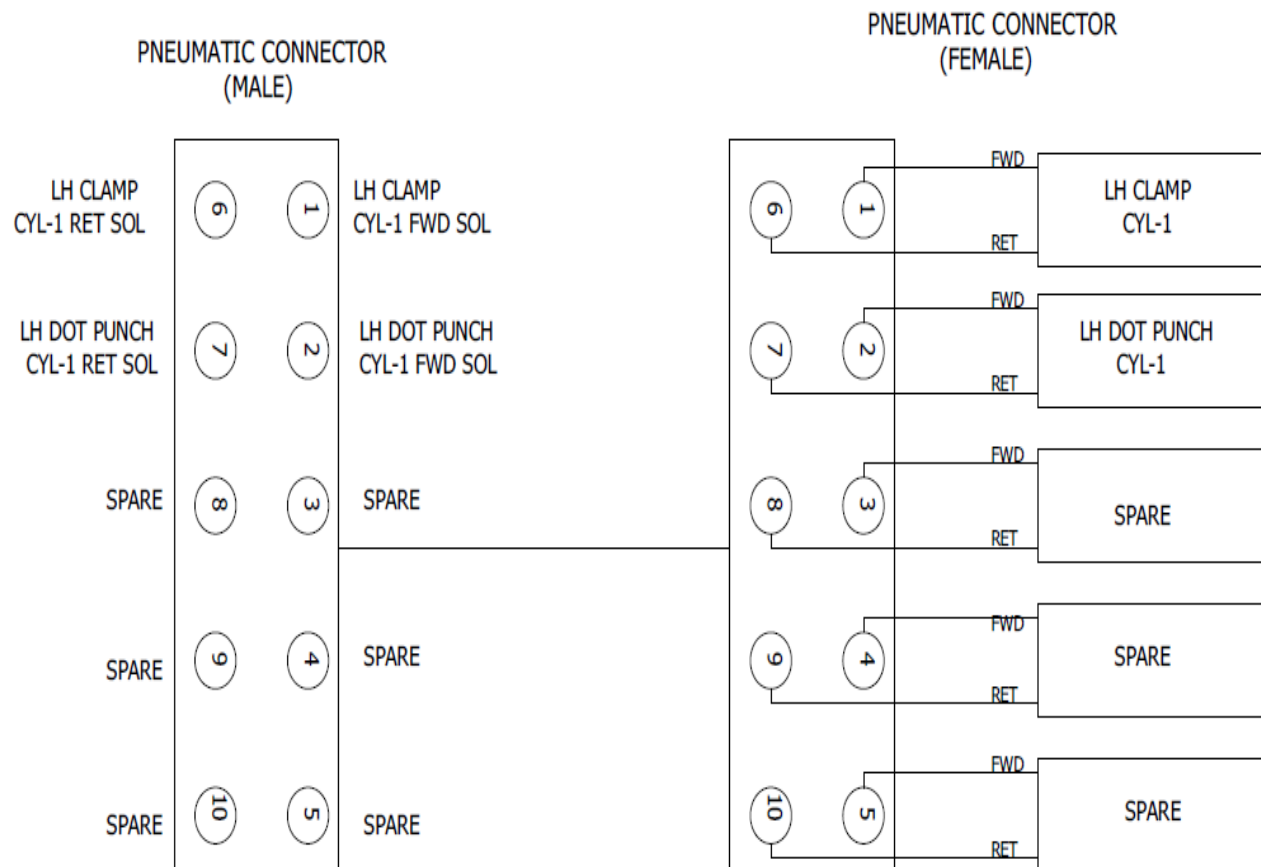
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LH SIDE PNEUMATIC LAYOUT

LH SIDE - PNEUMATIC CONNECTOR





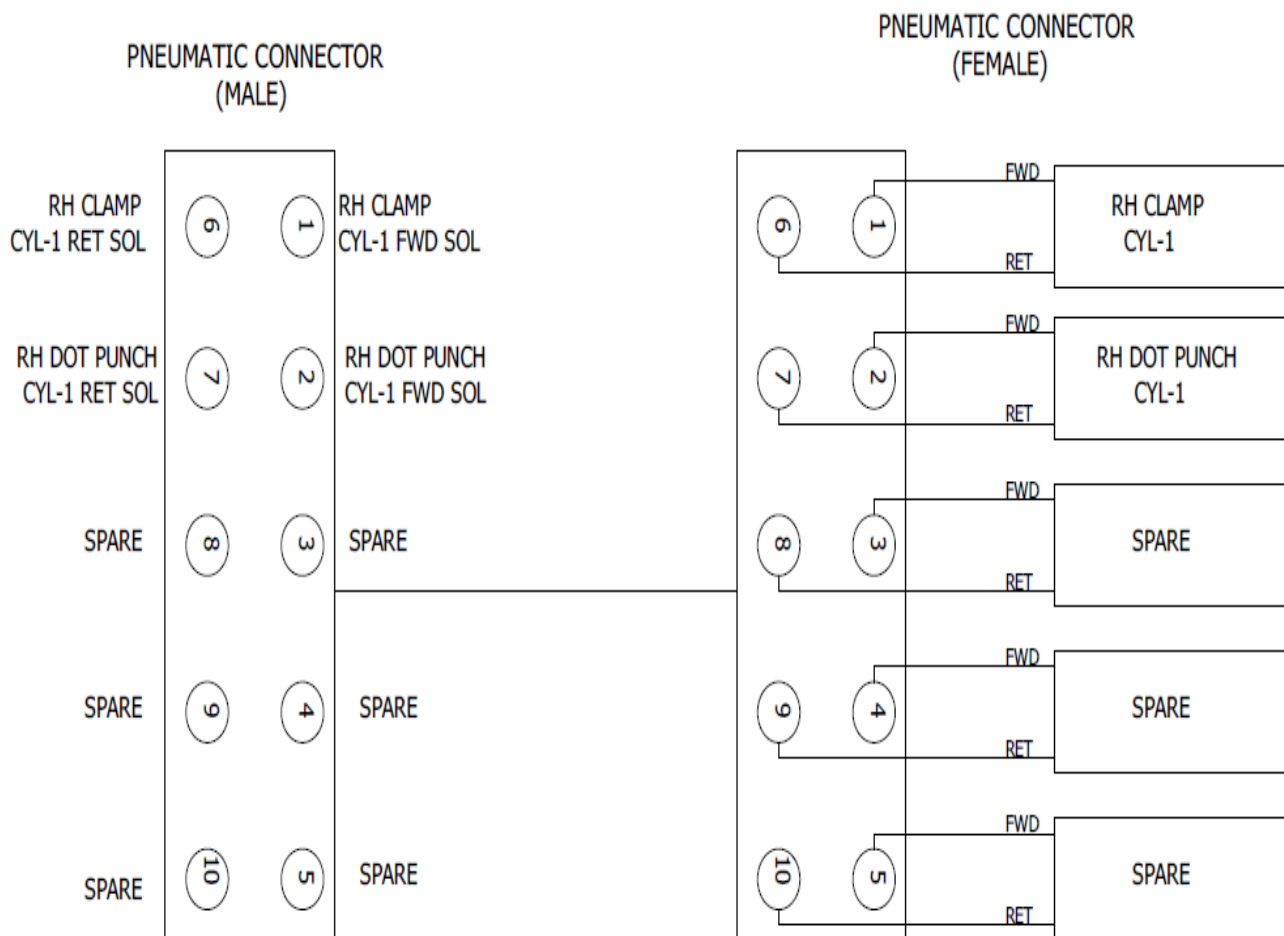
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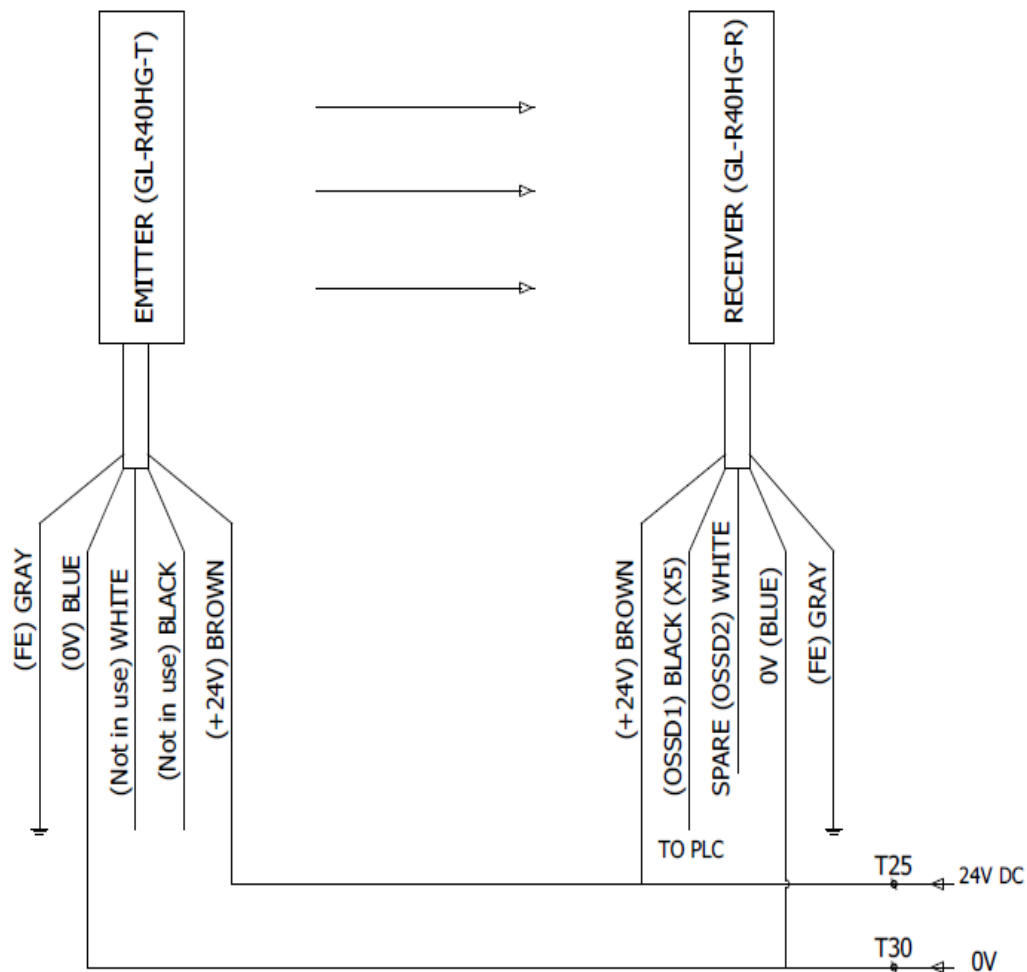
RH SIDE PNEUMATIC LAYOUT

RH SIDE - PNEUMATIC CONNECTOR



LIGHT CURTAIN CONNECTIONS (KEYENCE: -GL-R36-HG)

LIGHT CURTAIN CONNECTIONS (KEYENCE:-GL-R36-HG)





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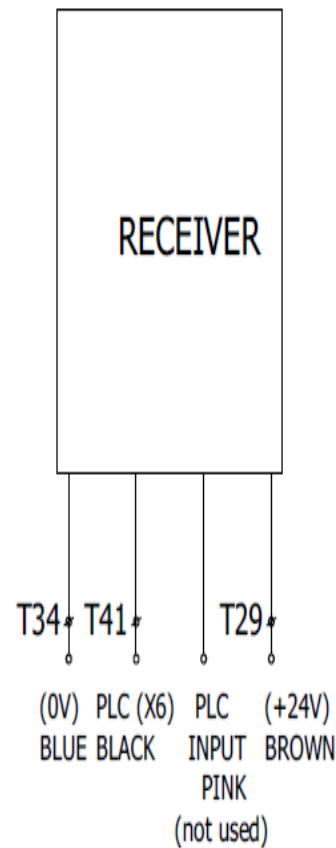
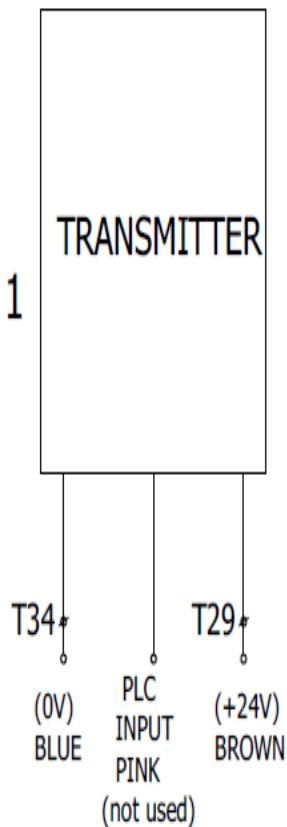
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REJECTION BIN 1 CONNECTIONS (PANASONIC: -NA1-5-PN)

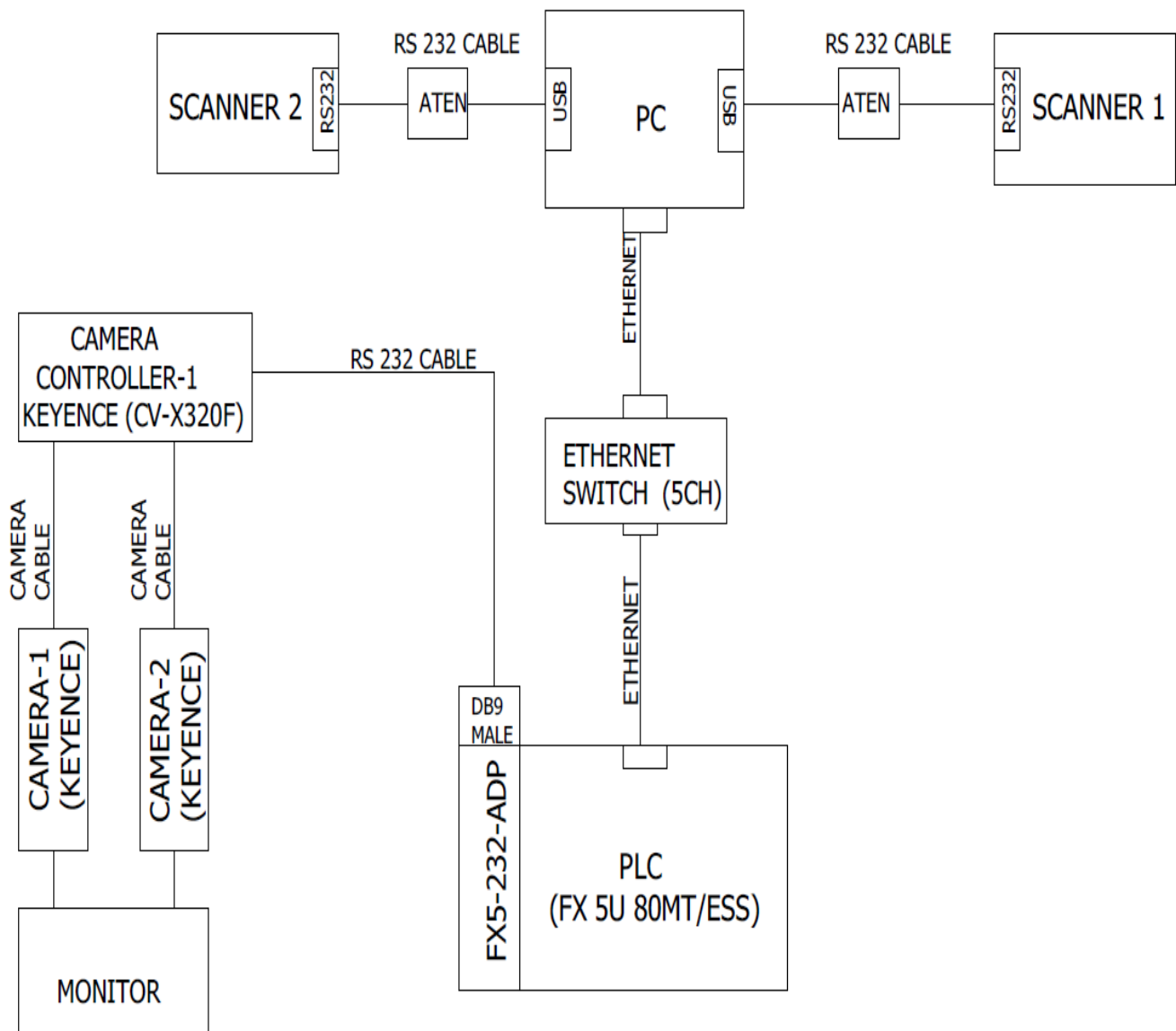
REJECTION BIN 1 CONNECTIONS (PANASONIC:-NA1-5-PN)

REJECTION
BIN SENSOR 1



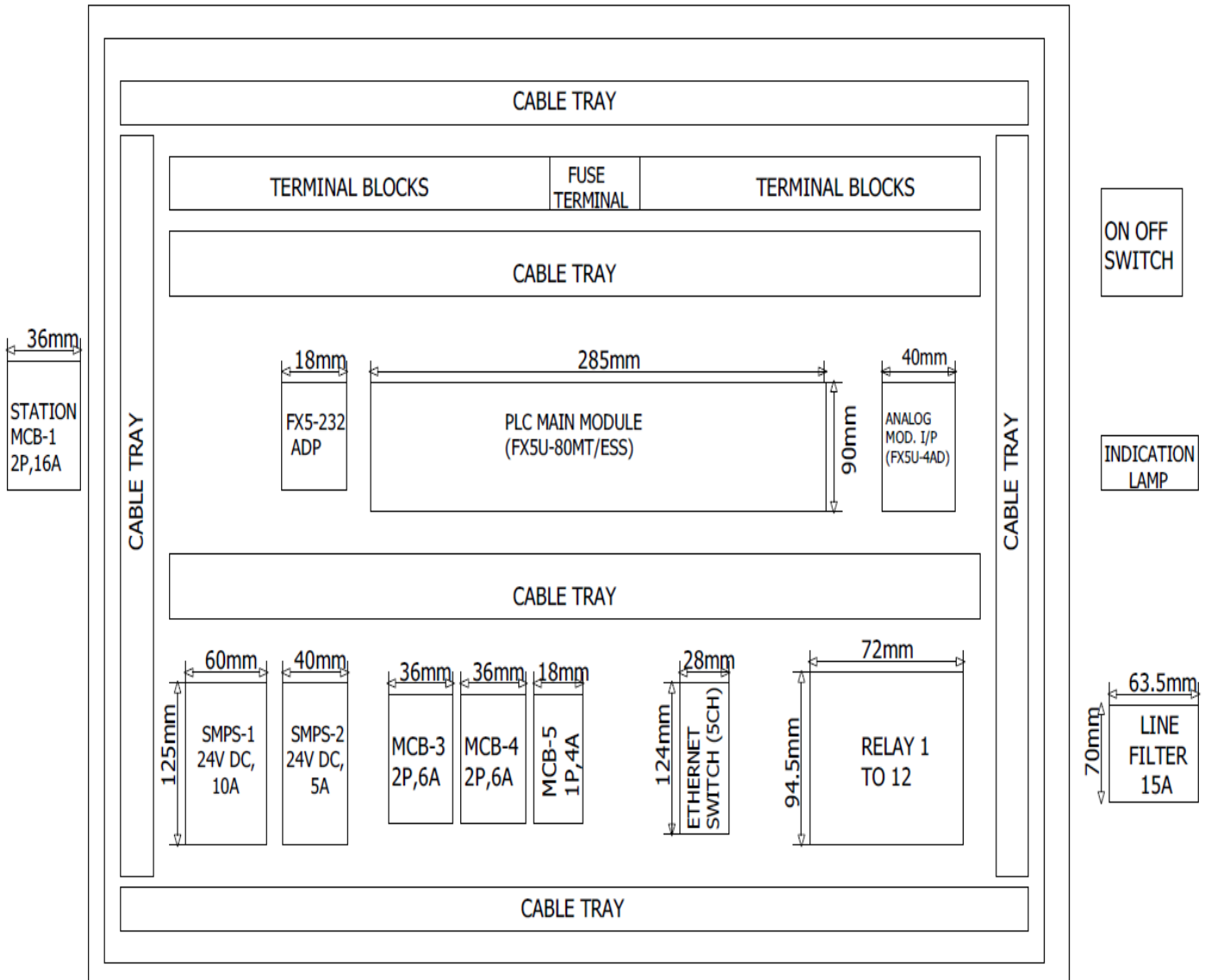
COMMUNICATION ARCHITECTURE

COMMUNICATION ARCHITECTURE



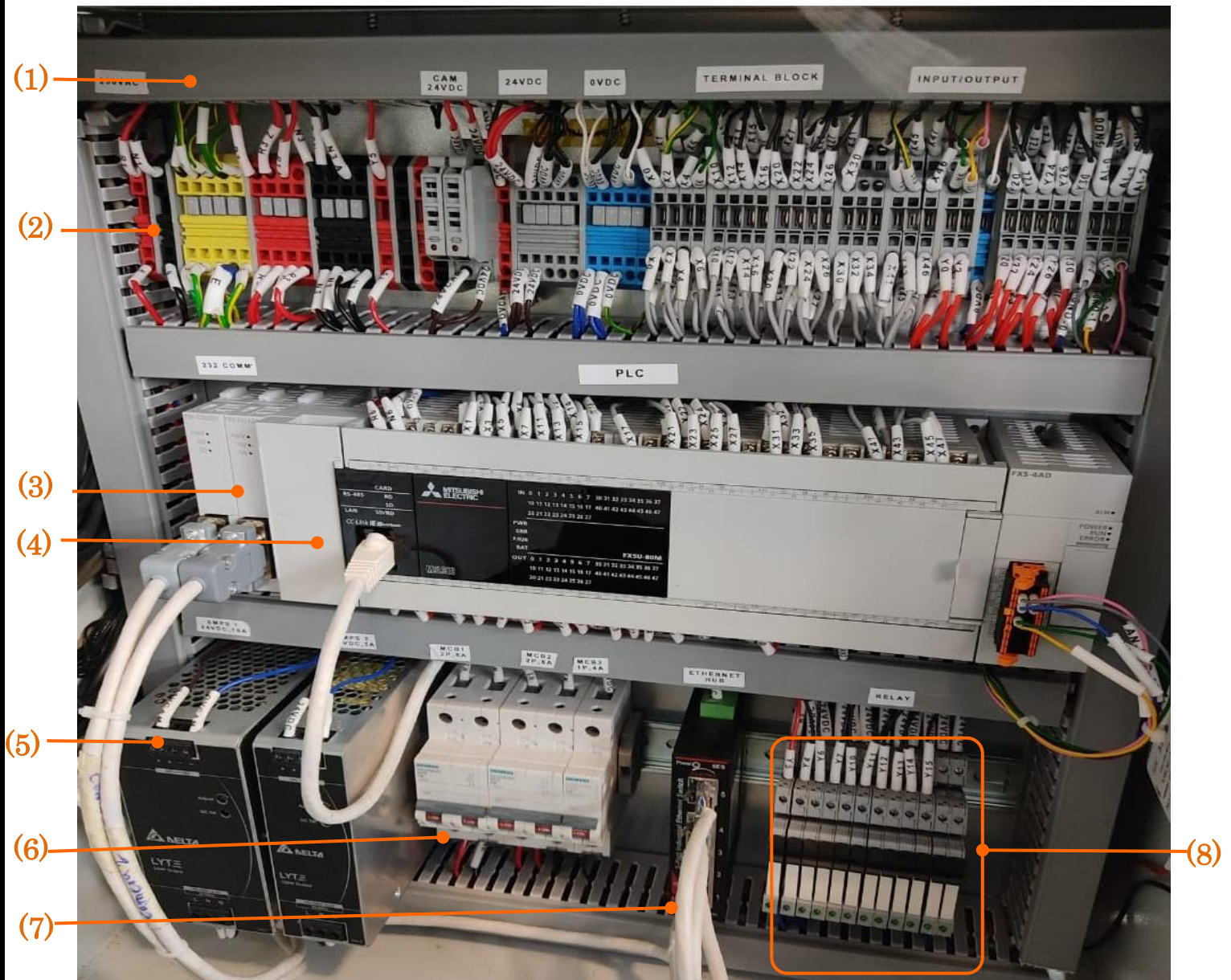
COMPONENT LAYOUT

COMPONENT LAYOUT



Component Layout

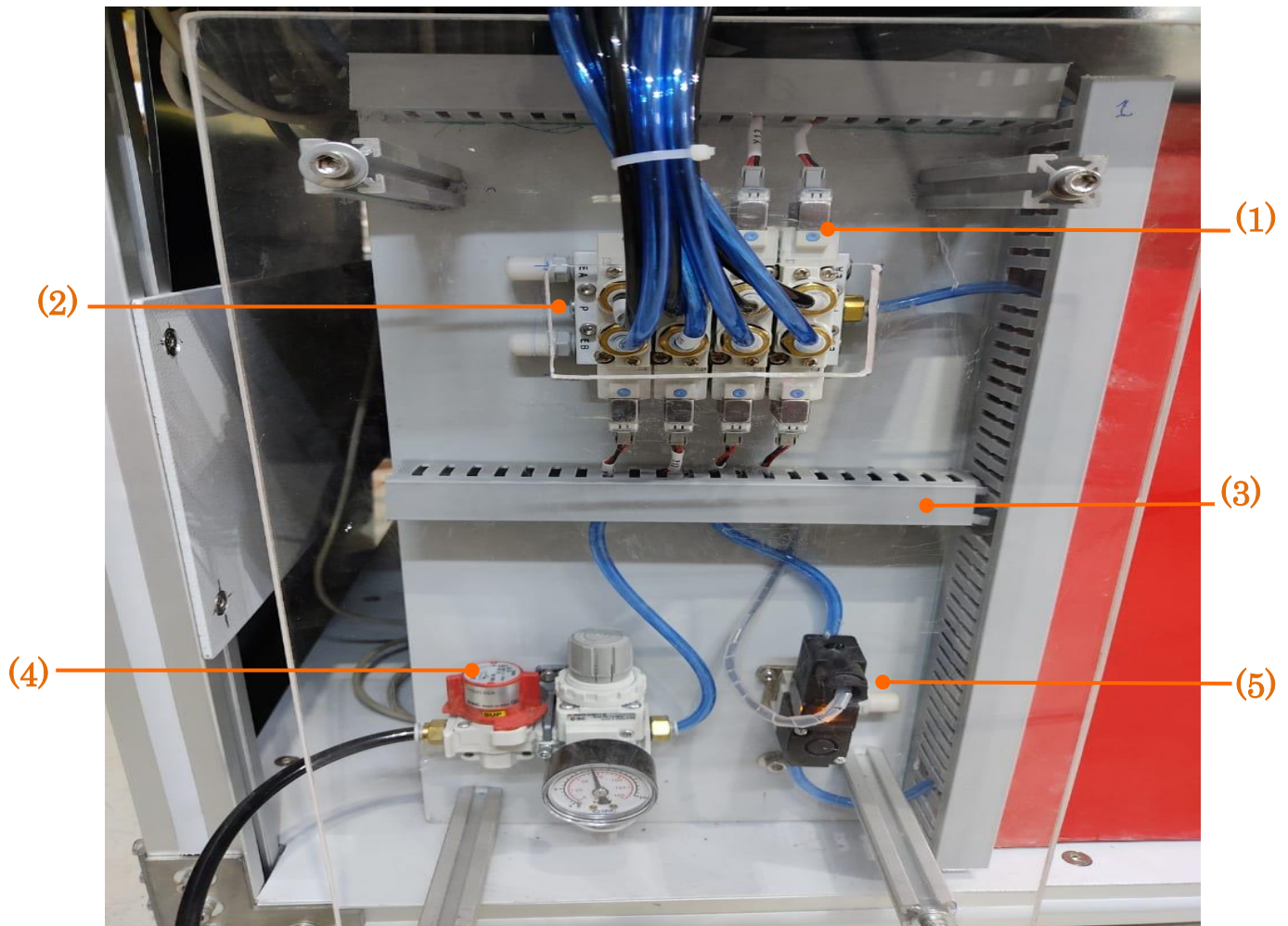
Internal Panel Layout and Pneumatic Plate Details



No.	Component Name	Description
1	Cable Tray	To route internal wires.
2	Terminal Blocks	To connect external devices.

3	FX5-232ADP	RS232 communication module.
4	PLC	To input output controller.
5	SMPS (24 V, 2.5 A)	Used for AC to DC converter.
6	MCB	To Protection against short circuit and overloading current.
7	Ethernet switch	5 channel communication port
8	1-channel Relays	To connect solenoid valves.

Pneumatic Plate Details





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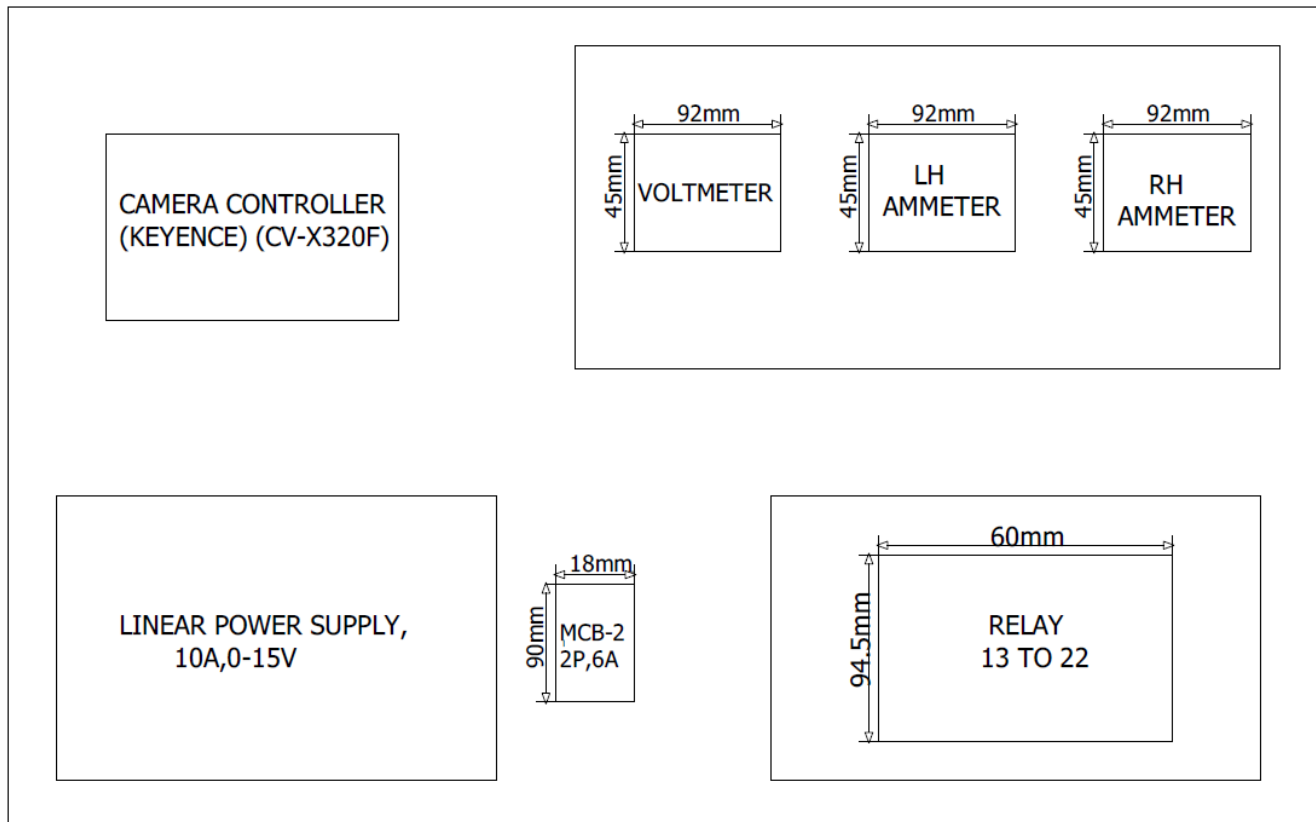
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No.	Component Name	Description
1	Double acting solenoid valve	For cylinder operation.
2	Manifold	Suitable mounting of solenoid valve.
3	Cable tray	To internal wires.
4	FR Unit	Air filter & regulator unit.
5	Air off solenoid valve	To on/off air

FIELD LAYOUT

FIELD LAYOUT





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BILL OF MATERIAL:

BOM

ELECTRICAL PANEL MATERIAL

S N	COMPONENT NAME	DESCRIPTION	PART NO.	MAKE	QTY.
1	Control Panel	500X500X210	MADO505020R5	Mfg	1
2	PLC	40DI,40DO	FX5U-80MT/ESS	Mitsubishi	1
3	Analog Module	4DI	FX5-4AD	Mitsubishi	1
4	Expansion module RS-232		FX5-232ADP	Mitsubishi	1
5	Linear power supply	10A,0-15V		Ultrawatt	1
6	SMPS	24 VDC,10A	DRL-24V240W1AA	DELTA	1
7	SMPS	25 VDC,5A	DRL-24V120W1AA	DELTA	1
8	Push Button (Green)	Green Mushroom Type	2AM3	Teknic	2
9	Reset Button(red)	Red Flush Type	2AM4	Teknic	1
10	Emergency Stop	Latched	2AML4	Teknic	1
11	NO Element	24 VDC	S1	Teknic	3
12	NC Element	24 VDC	S2	Teknic	1
13	PB Box	2Way	50MM	Teknic	2
14	Indication Lamp (Green)	230 VAC	3PLBR-3L	Teknic	1
15	Relay	24VDC,6A	SOCY1-24D	Connectwell	22
16	ON/Off Switch	2P,16A,230VAc	61002SAB13TDYR	L&t	1
17	Rejection bin		NA1-5-PN	Panasonic	4
18	MCB (Zeal input)	2P,6A	5SL62067RC	Simens	3
19	MCB (Pannel)	2P,16A	5SL42167RC	Simens	1
20	MCB (24VDC)	1P,4A	5SL61047RC	Simens	1
21	Tower Lamp 3-Tier	Steady,24 VDC	MG-302B-RYG	Patellite	1
22	Fuse terminal	4A		Wago	5
23	Electric indo	24 Pin Regular, male female with housing		Indo	2
24	Panel fan	230VAC	4"	Super Flow	1
25	LED Tube Light	230VAC		SS Light	2

	YFG FCL LIGHT TEST.	
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26	Extension Board &sw	6A(Monitor-1 , spare-2)	6755 50	Legrand	2
27	Station Fan	230VAC		ALMANRO	1
28	Ammeter	230V,AC	MT4W-DA	Autonics	2
29	Voltmeter	230V,AC	MT4W-DV	Autonics	1
30	Light curtain		GL-R36-HG	Keyence	1
31	Industrial socket	230v,3P,16A		Legrand	1
32	Emi emc filter	15A single phase		Elcom	1
33	Scanner			Datalogic	2
34	Serial to usb			Aten	4
35	Camera controller		CV-X320F	Keyence	1
36	Camera		CA-200C	Keyence	2
37	Camera cable			Keyence	2
38	MONITER + VGA CABLE			LG	1
39	Sensors				0
40	Ethernet hub			Sixnet	1
41	Computer & monitor	PC Configuration			1
		i3 10th or higher			1
		8GB or higher Ram			1
		500 gb or higher SSD			
		USBPort - 6 or higher			
		Monitor 21inch HDMI			
		Keyboard			
		Mouse			
		Windows10			
		UPS			
		Office(Excel)			

YFG FCL LIGHT TEST					
Sr No	Component Name	Description	Part No.	Make	Qty.
1	Solenoid Valve	Double Acting	SY7227-5LZ-02	SMC	2
		Single Acting	SY7120-5LZ-02	SMC	2
2	Read switch	24VDC, PNP		SMC	10
3	Air Off Solenoid Valve	0.7MPa	VT307-5DZ1-02	SMC	1
4	digital pressure switch	Pressure Display	ISE30AF-01-P-LA2	SMC	1
5	FRL unit	1/4"	AW20-02BG1-B	SMC	1
6	Quick Coupling connector 10 pin	Male+female	KDM10S-06	smc	2
7	Manifold	4 Way manifold	SS5Y7-20-06	SMC	1



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I/O LIST:

YFG FCL LIGHT TEST					
MODULES	I/P ADDR	INPUTS	O/P ADDR	OUTPUTS	RELAY
FX5U-80MT/ESS	X0	START PB 1	Y0	PASS LAMP	
	X1	START PB 2	Y1	BUZZER	
	X2	EMB PB	Y2	PROCESS LAMP	
	X3	RESET	Y3	ERROR LAMP	
	X4	AIR PRESSURE SWITCH	Y4	AIR OFF SOL	R1
	X5	LIGHT CURTAIN	Y5	PANEL FAN	R2
	X6	SPARE	Y6	SPARE	R3
	X7	SPARE	Y7	SPARE	R4
	X10	LH CLAMP FWD REED	Y10	LH CLAMP FWD SOL	R5
	X11	LH CLAMP RET REED	Y11	LH CLAMP RET SOL	R6
	X12	LH PUNCH FWD REED	Y12	LH PUNCH FWD/RET SOL	R7
	X13	LH PUNCH FWD REED	Y13	RH CLAMP FWD SOL	R8
	X14	LH NITTO CAP 1	Y14	RH CLAMP RET SOL	R9
	X15	LH NITTO CAP 2	Y15	RH PUNCH FWD/RET SOL	R10
	X16	LH NITTO CAP 3	Y16	SPARE	R11
	X17	LH NITTO CAP 4	Y17	SPARE	R12
	X20	LH PART PRESENT SENSOR 5	Y20	LH LAMP 1 (TURN)	R13
	X21	RH CLAMP FWD REED	Y21	LH LAMP 2 (POS)	R14
	X22	RH CLAMP RET REED	Y22	LH LAMP 3 (DRL)	R15
	X23	RH PUNCH FWD REED	Y23	LH LAMP 4	R16
	X24	RH PUNCH RET REED	Y24	LH LAMP 5	R17
	X25	RH NITTO CAP 1	Y25	RH LAMP 1 (TURN)	R18
	X26	RH NITTO CAP 2	Y26	RH LAMP 2 (POS)	R19
	X27	RH NITTO CAP 3	Y27	RH LAMP 3 (DRL)	R20
	X30	RH NITTO CAP 4	Y30	RH LAMP 4	R21
	X31	RH PART PRESENT SENSOR 5	Y31	RH LAMP 5	R22
	X32	SHUTTER RET RD	Y32	SPARE	
	X33	SHUTTER FWD RD	Y33	SPARE	
	X34	LH PART PRESENT SENSOR 6	Y34	SPARE	
	X35	RH PART PRESENT SENSOR 6	Y35	SPARE	
	X36	REJECTION BIN SENSOR-1	Y36	SPARE	
	X37	REJECTION BIN SENSOR-2	Y37	SPARE	
	X40	REJECTION BIN SENSOR-3	Y40	LEAK PROGRAM SEL 1	
	X41	LH FIXTURE SEL-1	Y41	LEAK PROGRAM SEL 2	
	X42	LH FIXTURE SEL-2	Y42	LEAK PROGRAM SEL 3	
	X43	LH FIXTURE SEL-3	Y43	LEAK PROGRAM SEL 4	



YFG FCL LIGHT TEST.

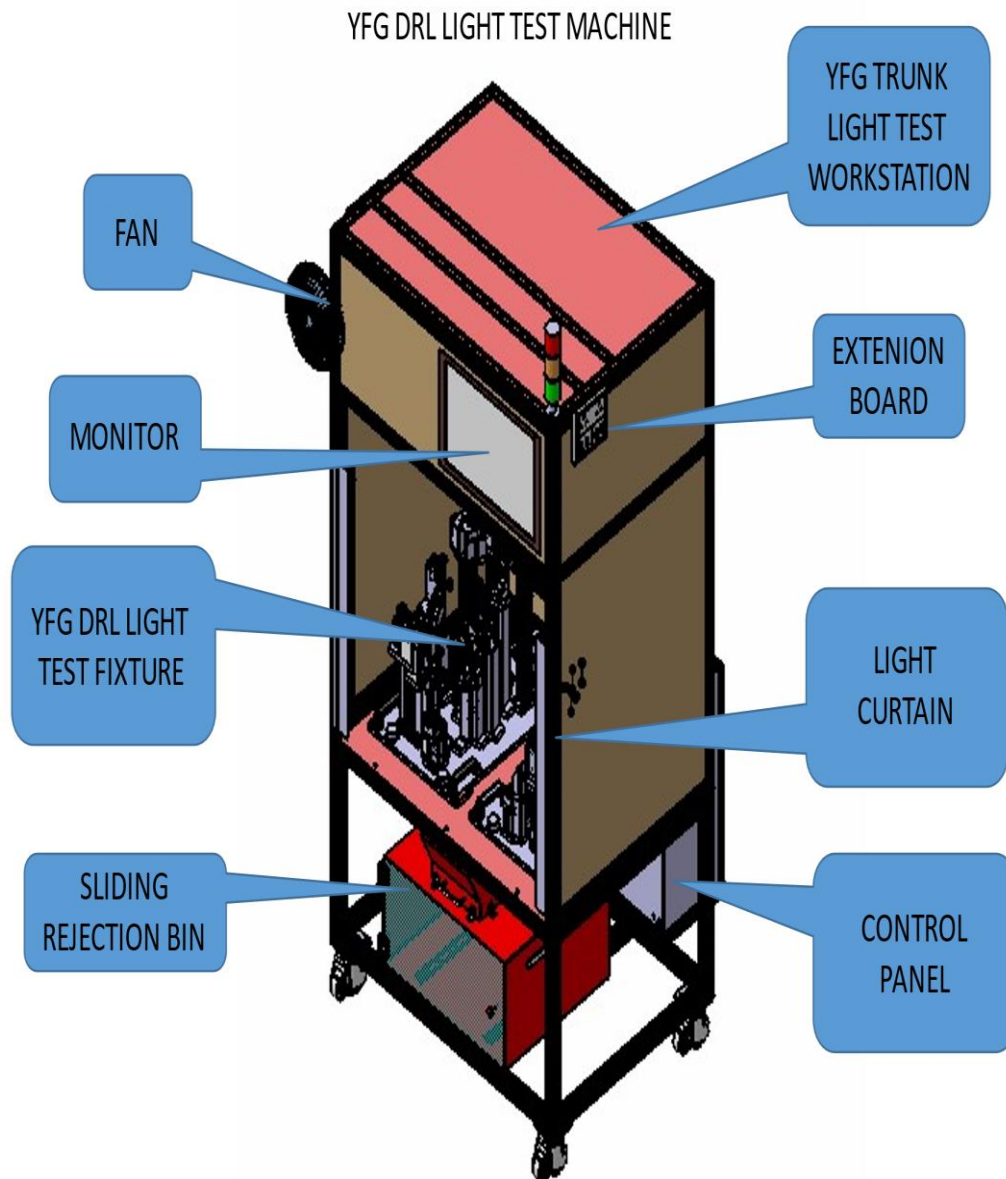
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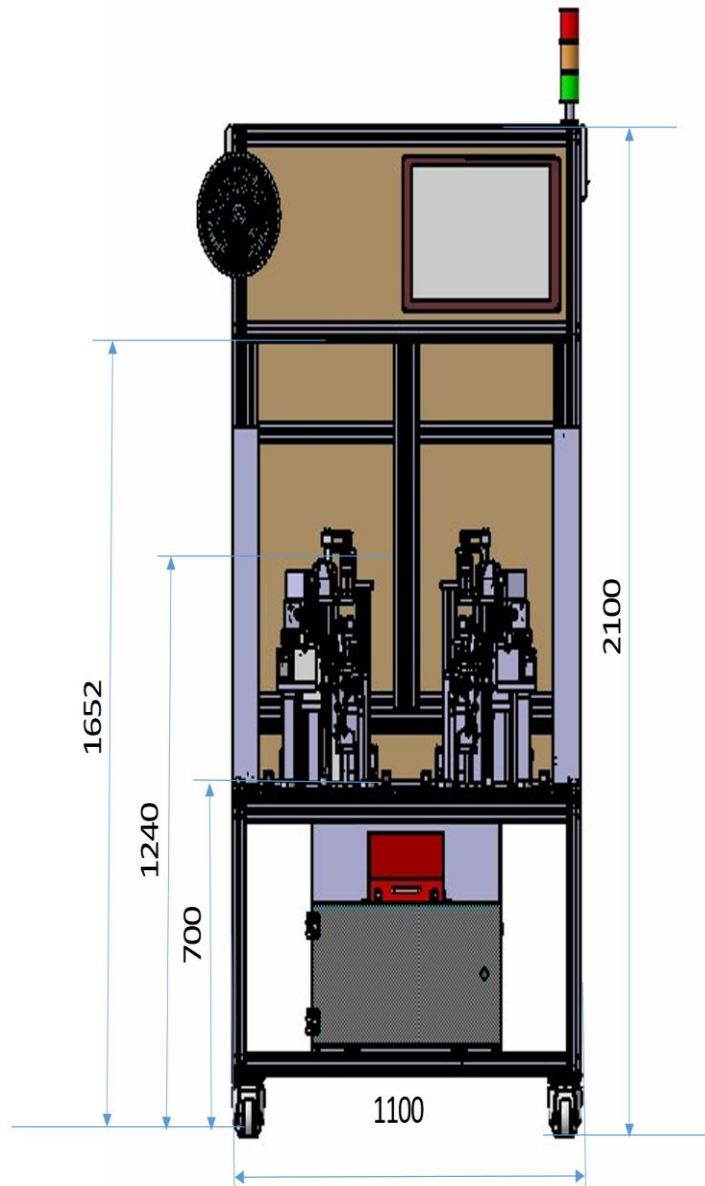
	X44	REJECTION BIN SENSOR-4	Y44	SPARE	
	X45	RH FIXTURE SEL-1	Y45	SPARE	
	X46	RH FIXTURE SEL-2	Y46	SPARE	
	X47	RH FIXTURE SEL-3	Y47	SPARE	
FX5-4AD	AI-X0	LAMP VOLTAGE			
	AI-X1	LH LAMP CURRENT			
	AI-X2	RH LAMP CURRENT			
	AI-X3	SPARE			
COMMUNICATION	PLC - VB	ETHERNET PORT			
	VB - SCANNER1	SERIAL TO USB (ATEN)			
	VB - SCANNER2	SERIAL TO USB (ATEN)			
	PLC- CAMERA 1	FX5-232-ADP			
	PLC- CAMERA 2	FX5-232-ADP			

MECHANICAL DETAILS:

COMPONENT DESCRIPTION

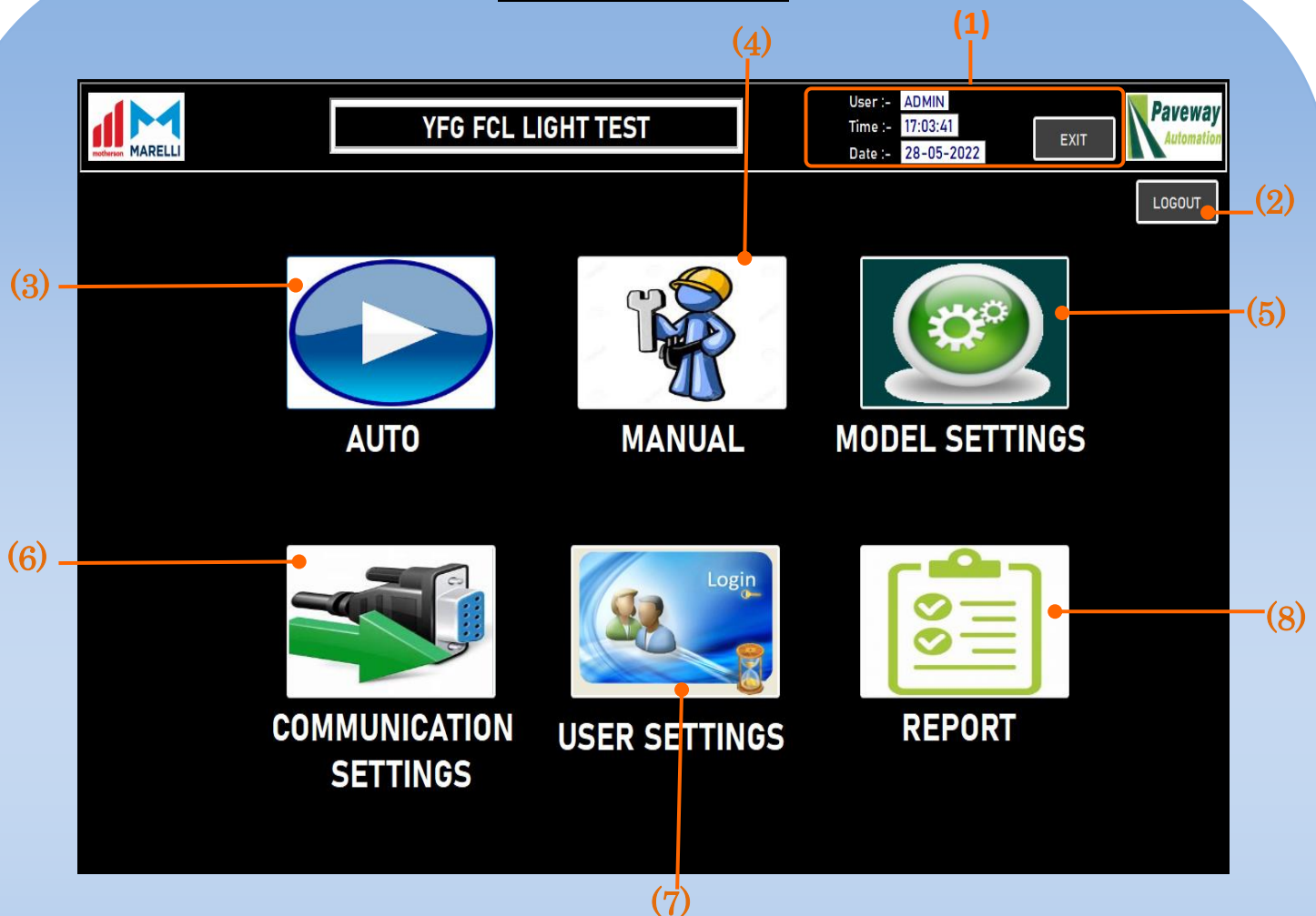


OVERALL DIMENSION



VB SCREEN DETAILS:

HOME SCREEN



1.Program name, Date & Time display. 6.Communication setting screen navigation button.

- Exit navigation button.

7.User setting screen navigation button.

2. Logout navigation button.

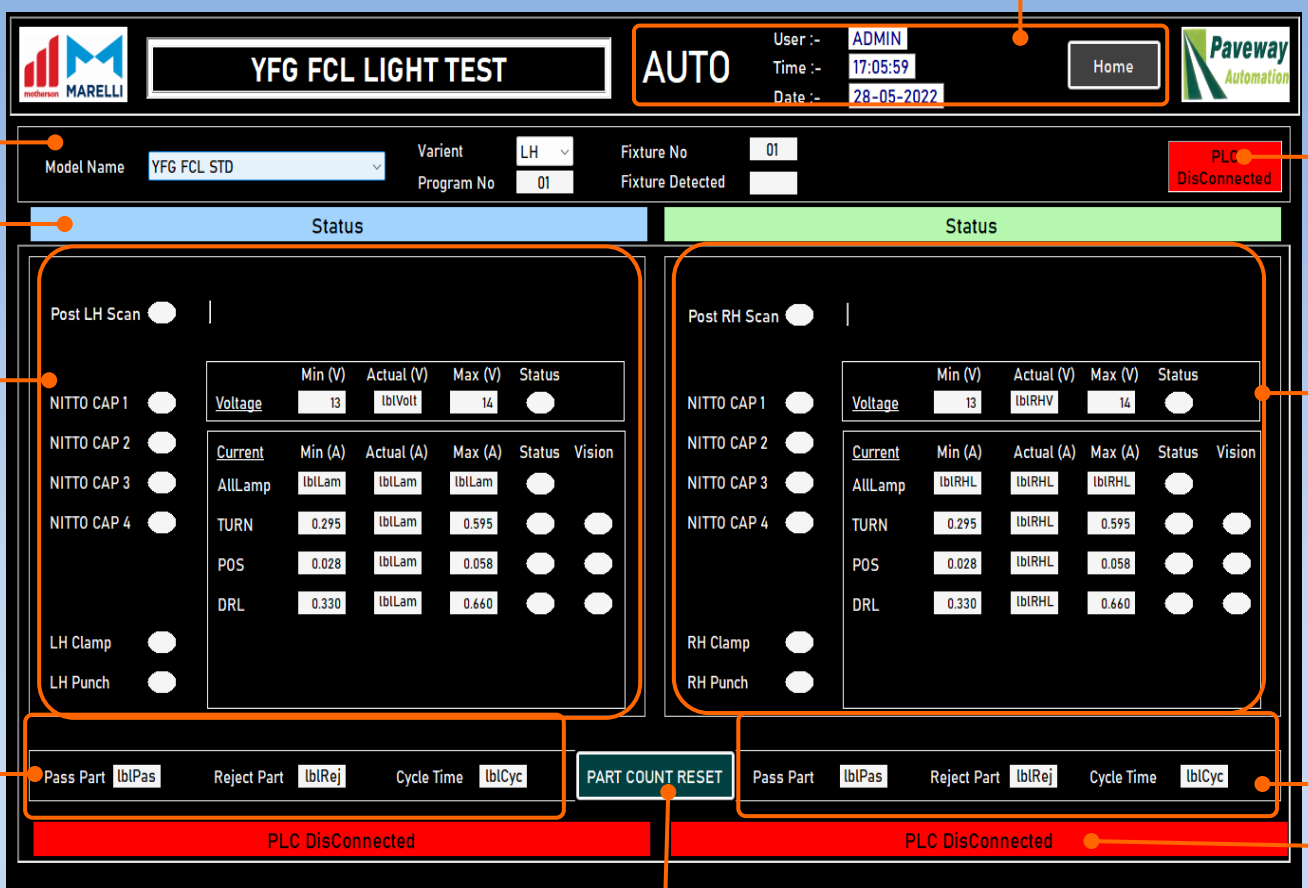
8.Report screen navigation button.

3.Auto mode screen navigation button.

4.Manual mode screen navigation button.

5.Model setting screen navigation button.

AUTO MODE SCREEN-1



The screenshot shows the 'YFG FCL LIGHT TEST' interface in 'AUTO' mode. It includes a header with user information (ADMIN, 17:05:59, 28-05-2022) and a 'Home' button. Below the header, there are fields for Model Name (YFG FCL STD), Variant (LH), Program No (01), Fixture No (01), and Fixture Detected status. A red 'PLC DisConnected' indicator is visible. The main area is divided into two columns for 'Status' messages and 'Post LH Scan'/'Post RH Scan' status displays. Each scan display includes a table with parameters like Voltage, Current, and Turn, along with their minimum, actual, and maximum values. At the bottom, there are fields for Pass Part (lbiPas), Reject Part (lbiRej), Cycle Time (lbiCyc), and a 'PART COUNT RESET' button. A red bar at the very bottom indicates 'PLC DisConnected'.

1.Home screen navigation button.

- User name, date and time display.

2.Model name, variant and program number, fixture and fixture detected status display.

3.Status message display.

4.Post LH scan status display.

- Nitto cap 1,2,3,4 status display,
- LH clamp, LH punch status display.
- Voltage-minimum, actual, maximum parameter status display.



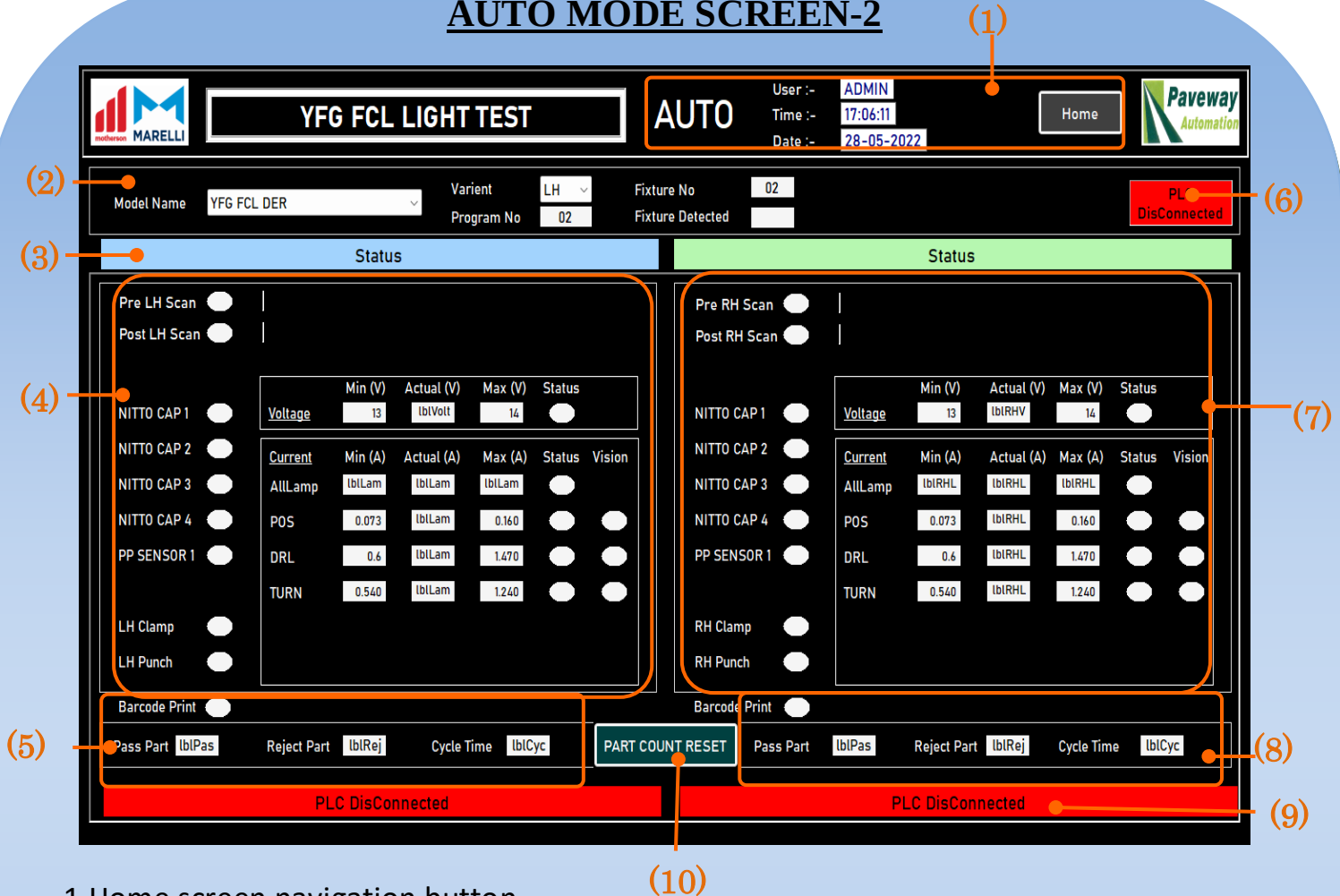
YFG FCL LIGHT TEST.

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- Current-Minimum, Actual, maximum parameter status display.
5. Pass part, Reject part, cycle time operation status display.
6. PLC disconnected error display.
7. Post RH scan status display.
- Nitto cap 1,2,3,4 status display,
 - LH clamp, LH punch status display.
 - Voltage-minimum, actual, maximum parameter status display.
 - Current-Minimum, Actual, maximum parameter status display.
8. Pass part, Reject part, cycle time operation status display.
9. Error message display.
10. Part count reset navigation button.

AUTO MODE SCREEN-2



The screenshot shows the 'YFG FCL LIGHT TEST' interface in 'AUTO' mode. The interface is divided into several sections:

- Header:** Includes the Paveway Automation logo, the title 'YFG FCL LIGHT TEST', the 'AUTO' mode indicator, and user information (User: ADMIN, Time: 17:06:11, Date: 28-05-2022). A 'Home' button is also present.
- Model Information:** Displays 'Model Name: YFG FCL DER', 'Variet: LH', 'Program No: 02', and 'Fixture No: 02'. A 'Fixture Detected' status is shown as 'DisConnected'.
- Status Messages:** Two 'Status' bars are visible, one for the left (LH) and one for the right (RH) side.
- Scan Status:** Sections for 'Pre LH Scan', 'Post LH Scan', 'Pre RH Scan', and 'Post RH Scan' are shown.
- Parameter Tables:** Two tables display test parameters for 'NITTO CAP 1', 'NITTO CAP 2', 'NITTO CAP 3', 'NITTO CAP 4', 'PP SENSOR 1', 'LH Clamp', and 'LH Punch' (or 'RH' for the right side). Each table includes columns for 'Min (V)', 'Actual (V)', 'Max (V)', and 'Status'.
- Barcode Print:** A button to print barcode information.
- Pass/Reject/Cycle Time:** Fields for 'Pass Part', 'Reject Part', and 'Cycle Time' are displayed.
- PLC DisConnected:** A red bar at the bottom indicates the PLC status.
- PART COUNT RESET:** A button to reset the part count.

1.Home screen navigation button.

- User name, date and time display.

2.Model name, variant and program number, fixture and fixture detected status display.

3.Status message display.

4.Pre and Post LH scan status display.

- Nitto cap 1,2,3,4 status display,
- LH clamp, LH punch status display.
- Voltage-minimum, actual, maximum parameter status display.



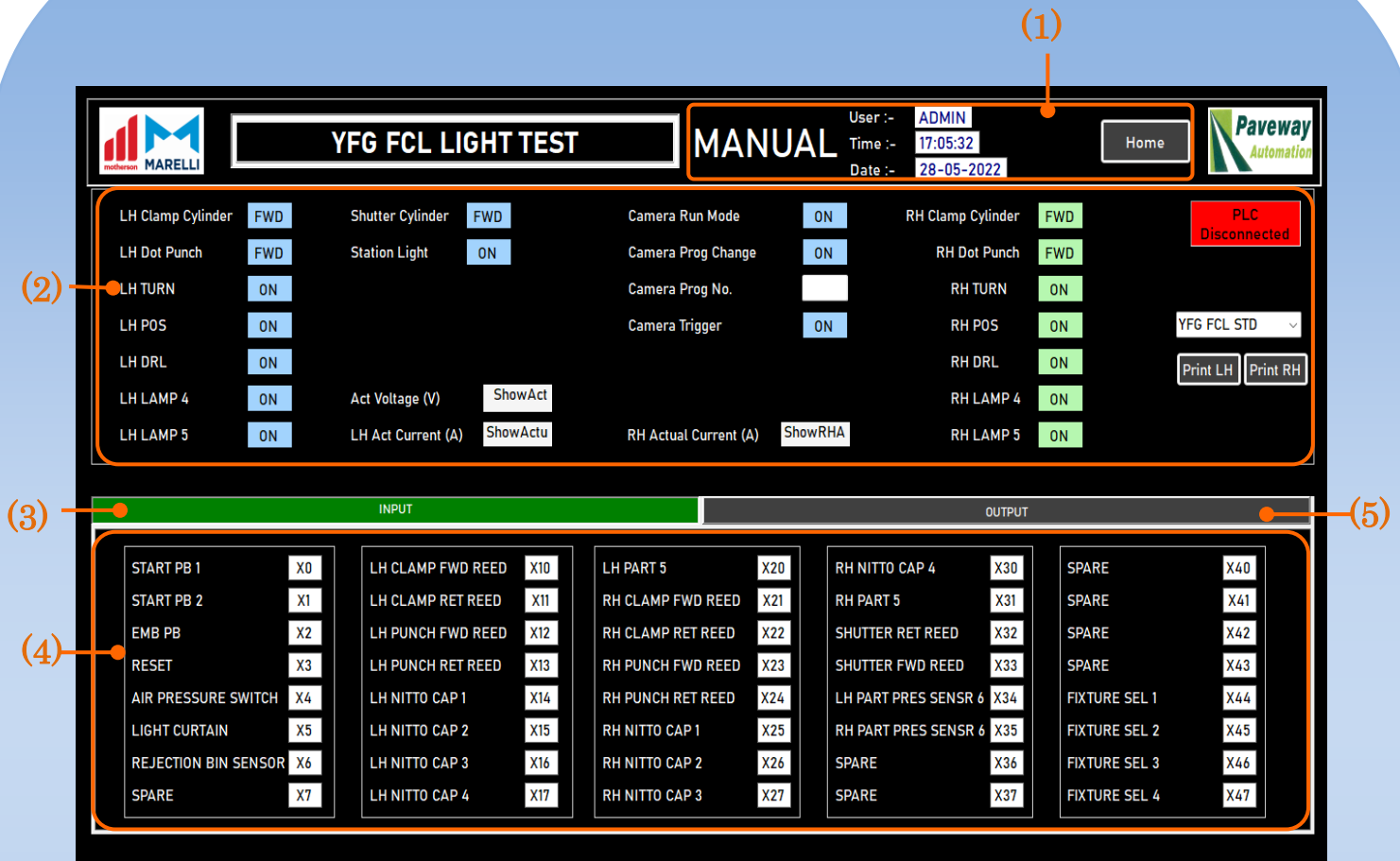
YFG FCL LIGHT TEST.

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- Current-Minimum, Actual, maximum parameter status display.
5. Pass part, Reject part, cycle time operation status display.
6. PLC disconnected error display.
7. Pre and Post RH scan status display.
- Nitto cap 1,2,3,4 status display,
 - LH clamp, LH punch status display.
 - Voltage-minimum, actual, maximum parameter status display.
 - Current-Minimum, Actual, maximum parameter status display.
8. Pass part, Reject part, cycle time operation status display.
9. Error message display.
10. Part count reset navigation button.

MANUAL MODE SCREEN-1



1.Home screen navigation button.

- User name, date and time display.

2.LH clamp, Dot punch cylinder manually press fwd/ret operation status display.

- LH TURN, LH DRL, LH POS, LH lamp4, LH lamp 5 manually press on/off operation status display.actual voltage and actual current display.
- Shutter cylinder manually press fwd/ret operation status display.
- Station light manually on/off operation status display.
- Camera mode manually press on/off operation status display.



YFG FCL LIGHT TEST.

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- RH clamp, Dot punch cylinder manually press fwd/ret operation status display.
- RH TURN, RH DRL, RH POS, RH lamp4, RH lamp 5 manually press on/off operation status display.actual voltage and actual current display.
- PLC disconnected error display.
- Print LH and RH navigation button.

3. Input tab.

4. PLC input status.

5.Output tab.

MANUAL MODE SCREEN-2



YFG FCL LIGHT TEST
MANUAL

User :- **ADMIN**
Time :- **17:05:35**
Date :- **28-05-2022**

Home


LH Clamp Cylinder	FWD	Shutter Cylinder	FWD	Camera Run Mode	ON	RH Clamp Cylinder	FWD	PLC Disconnected
LH Dot Punch	FWD	Station Light	ON	Camera Prog Change	ON	RH Dot Punch	FWD	
LH TURN	ON			Camera Prog No.		RH TURN	ON	
LH POS	ON			Camera Trigger	ON	RH POS	ON	YFG FCL STD
LH DRL	ON					RH DRL	ON	Print LH Print RH
LH LAMP 4	ON	Act Voltage (V)	ShowAct			RH LAMP 4	ON	
LH LAMP 5	ON	LH Act Current (A)	ShowActu	RH Actual Current (A)	ShowRHA	RH LAMP 5	ON	

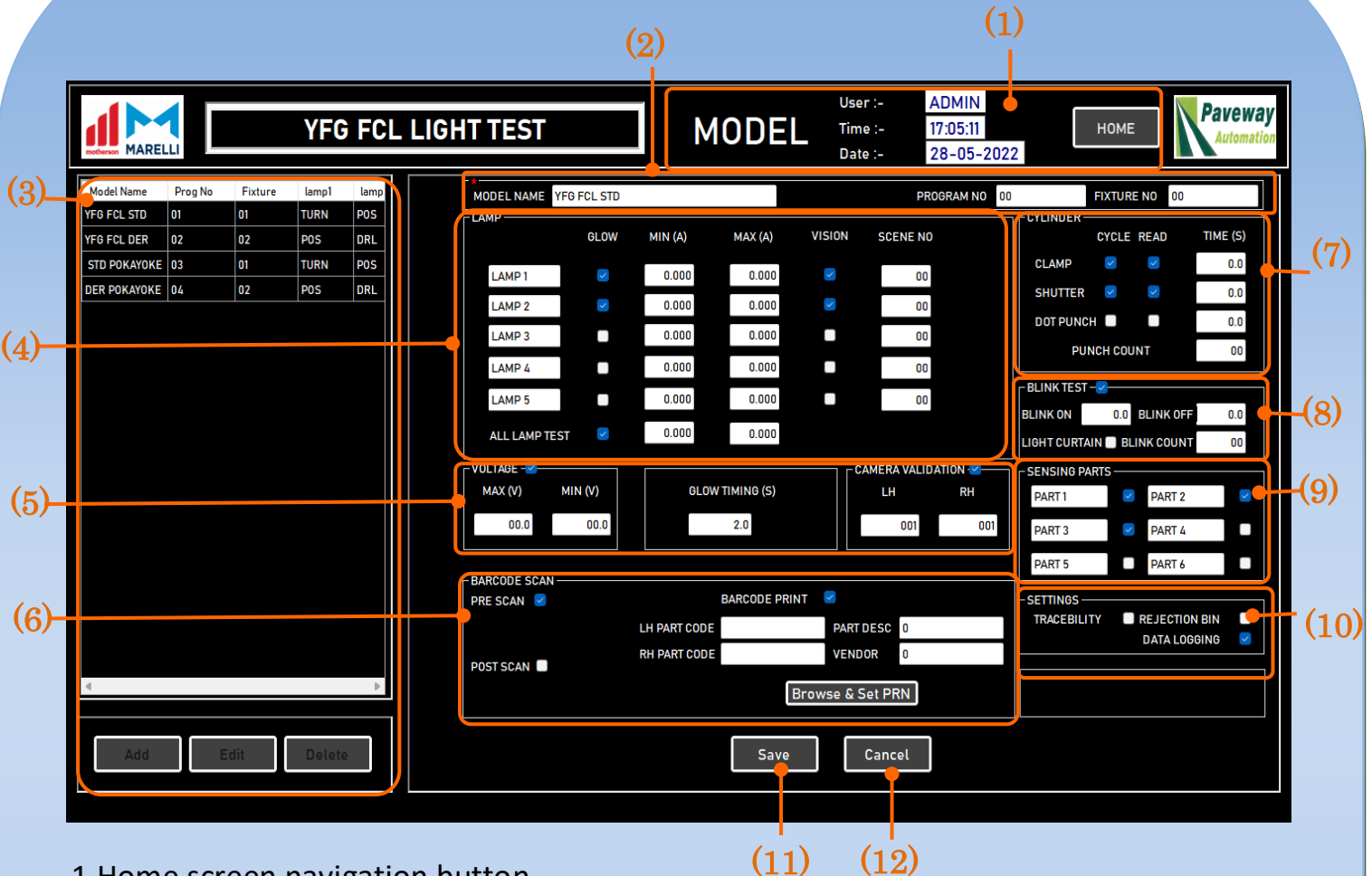
INPUT

OUTPUT

PASS LAMP	Y0	LH CLAMP FWD SOL	Y10	LH TURN	Y20	RH LAMP 4	Y30	LEAK PROG SEL1	Y40
BUZZER	Y1	LH CLAMP RET SOL	Y11	LH POS	Y21	RH LAMP 5	Y31	LEAK PROG SEL2	Y41
PROCESS LAMP	Y2	LH PUNCH FWD/RET SOL	Y12	LH DRL	Y22	SPARE	Y32	LEAK PROG SEL3	Y42
ERROR LAMP	Y3	RH CLAMP FWD SOL	Y13	LH LAMP 4	Y23	SPARE	Y33	LEAK PROG SEL4	Y43
AIR OFF SOL	Y4	RH CLAMP RET SOL	Y14	LH LAMP 5	Y24	SPARE	Y34	SPARE	Y44
PANEL FAN	Y5	RH PUNCH FWD/RET SOL	Y15	RH TURN	Y25	SPARE	Y35	SPARE	Y45
SPARE	Y6	SPARE	Y16	RH POS	Y26	SPARE	Y36	SPARE	Y46
SPARE	Y7	SPARE	Y17	RH DRL	Y27	SPARE	Y37	SPARE	Y47

1. Output tab.
2. PLC output status.

MODEL SETTING SCREEN-1



The screenshot shows the 'MODEL SETTING SCREEN-1' for 'YFG FCL LIGHT TEST'. The interface includes a header with the company logo and name, a user information section (User: ADMIN, Time: 17:05:11, Date: 28-05-2022), and a 'HOME' button. The main area is divided into several sections: a table of model names (YFG FCL STD, YFG FCL DER, STD POKAYOKE, DER POKAYOKE), a 'MODEL NAME' section with fields for 'MODEL NAME', 'PROGRAM NO', and 'FIXTURE NO', a 'LAMP' section with checkboxes for 'LAMP 1' through 'LAMP 5' and 'ALL LAMP TEST', a 'VOLTAGE' section with 'MAX (V)' and 'MIN (V)' fields, a 'GLOW TIMING (S)' field, a 'CAMERA VALIDATION' section with 'LH' and 'RH' fields, a 'CYLINDER' section with 'CYCLE', 'READ', and 'TIME (S)' fields, a 'BLINK TEST' section with 'BLINK ON' and 'BLINK OFF' fields, a 'LIGHT CURTAIN' section with 'BLINK COUNT' field, a 'SENSING PARTS' section with 'PART 1' through 'PART 6' checkboxes, and a 'SETTINGS' section with 'TRACEABILITY', 'REJECTION BIN', and 'DATA LOGGING' checkboxes. At the bottom, there are 'Add', 'Edit', 'Delete', 'Save', and 'Cancel' buttons. Numbered callouts (1) through (12) point to specific elements: (1) points to the 'HOME' button, (2) points to the 'MODEL NAME' section, (3) points to the model name list table, (4) points to the 'LAMP' section, (5) points to the 'VOLTAGE' section, (6) points to the 'CAMERA VALIDATION' section, (7) points to the 'CYLINDER' section, (8) points to the 'BLINK TEST' section, (9) points to the 'SENSING PARTS' section, (10) points to the 'SETTINGS' section, (11) points to the 'Save' button, and (12) points to the 'Cancel' button.

1.Home screen navigation button.

- User name, date and time display.

2.Model name, Program number and fixture number display.

3. Model name list display.

- Add, delete and edit navigation button.

4. Lamp settings:-

- Lamp-turn,pos,drl,lamp 4,lamp 5 select/unselect option and current in minimum and maximum setting.



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5. Voltage select/unselect option and parameter minimum and maximum setting.

- Glow timing display.
- Camera validation LH and RH select/unselect option.

6. Barcode scan setting-

- Pre scan,Pro scan select/unselect option.
- LH nad RH part code setting display.
- Browse and set PRN navigation button.

7. Cylinders reed switch select/unselect option and timing display.

8. Blink test parameter setting.

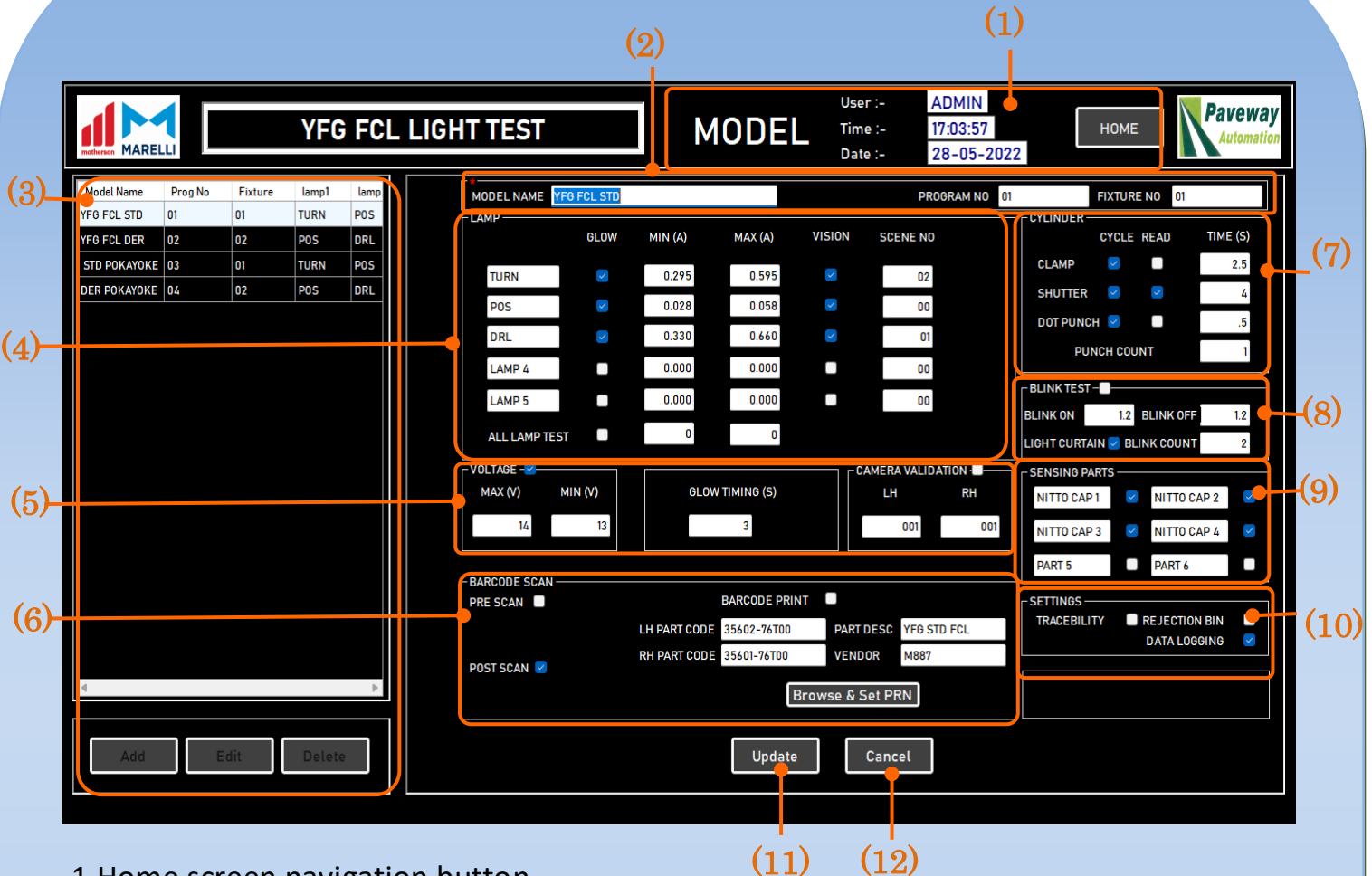
9. Sensing part nitto cap 1,2,3,4 select/unselect option.

10.Setting-Traceibility,Rejection bin and data logging select/unselect option.

11.Save navigation button.

12.Cancel navigation button.

MODEL SETTING SCREEN-2



The screenshot shows the 'MODEL SETTING SCREEN-2' for 'YFG FCL LIGHT TEST'. The interface includes a header with the company logo and name, a user information section (User: ADMIN, Time: 17:03:57, Date: 28-05-2022), and a 'HOME' button. The main area is divided into several sections: a table of model names (YFG FCL STD, YFG FCL DER, STD POKAYOKE, DER POKAYOKE), a 'MODEL NAME' dropdown (YFG FCL STD), 'PROGRAM NO' (01), and 'FIXTURE NO' (01). The 'LAMP' section contains a table with columns for LAMP, GLOW, MIN (A), MAX (A), VISION, and SCENE NO. The 'CYLINDER' section has a table with columns for CYLINDER, CYCLE, READ, and TIME (S). The 'BLINK TEST' section has a table with columns for BLINK ON, BLINK OFF, and BLINK COUNT. The 'SENSING PARTS' section has a table with columns for PART 1, PART 2, PART 3, PART 4, PART 5, and PART 6. The 'SETTINGS' section has a table with columns for TRACEABILITY, REJECTION BIN, and DATA LOGGING. The 'VOLTAGE' section has a table with columns for MAX (V), MIN (V), and GLOW TIMING (S). The 'CAMERA VALIDATION' section has a table with columns for LH and RH. The 'BARCODE SCAN' section has a table with columns for PRE SCAN, POST SCAN, LH PART CODE, RH PART CODE, PART DESC, and VENDOR. The 'UPDATE' and 'CANCEL' buttons are at the bottom.

1. Home screen navigation button.

2. Model name, Program number and fixture number display.

3. Model name list display.

4. Lamp settings:-

5. Add, delete and edit navigation button.

6. Lamp-turn,pos,drl,lamp 4,lamp 5 select/unselect option and current in minimum and maximum setting.

7. Cylinder settings.

8. Blink test settings.

9. Sensing parts settings.

10. Settings.

11. Update button.

12. Cancel button.



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5. Voltage select/unselect option and parameter minimum and maximum setting.

- Glow timing display.
- Camera validation LH and RH select/unselect option.

6. Barcode scan setting-

- Pre scan,Pro scan select/unselect option.
- LH nad RH part code setting display.
- Browse and set PRN navigation button.

7. Cylinders reed switch select/unselect option and timing display.

13.Blink test parameter setting.

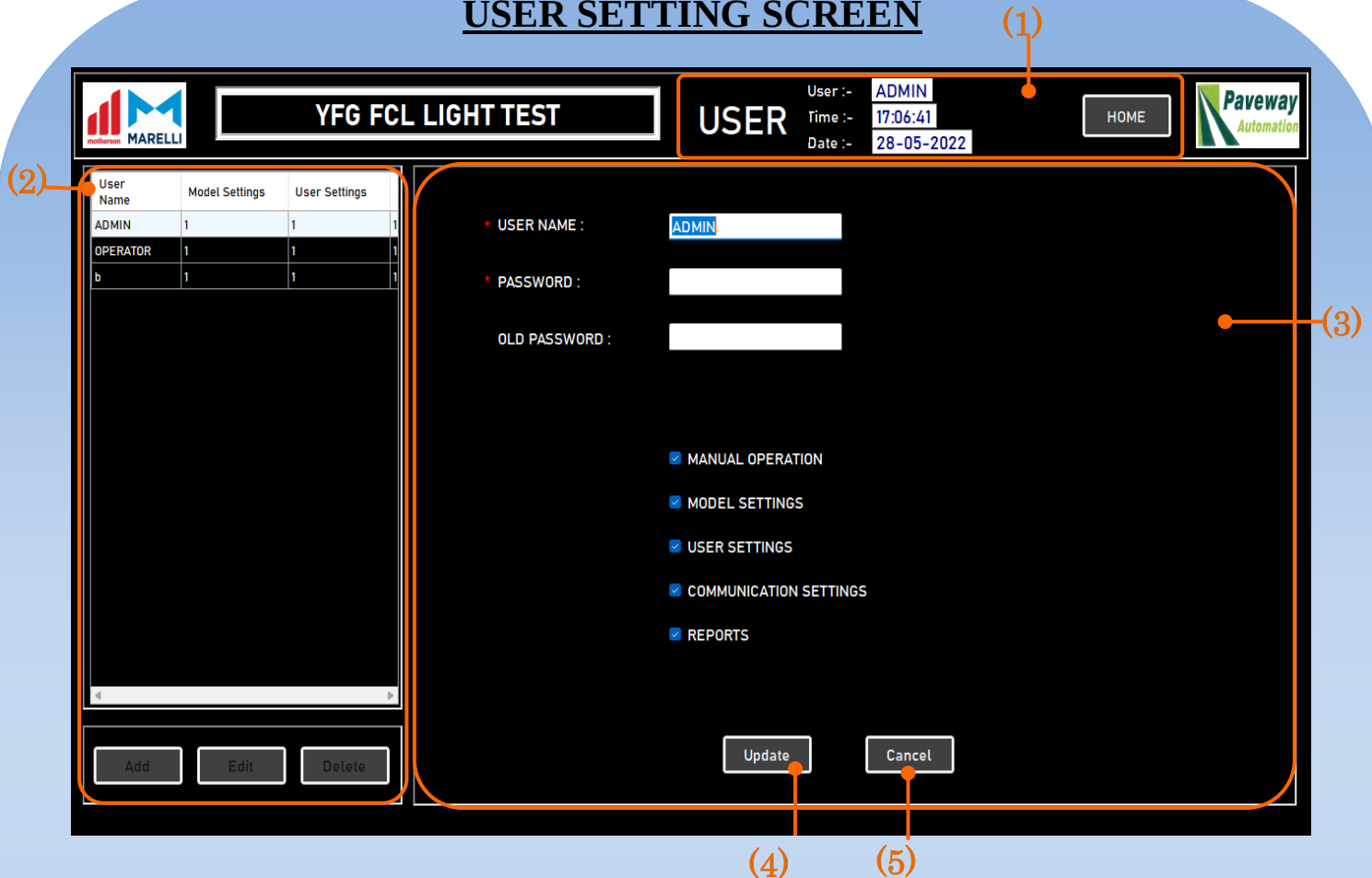
14.Sensing part nitto cap 1,2,3,4 select/unselect option.

15.Setting-Traceibility,Rejection bin and data logging select/unselect option.

16.Update navigation button.

17.Cancel navigation button.

USER SETTING SCREEN



The screenshot shows the 'USER SETTING SCREEN' interface. At the top, there is a header bar with the 'YFG FCL LIGHT TEST' title, a 'USER' section displaying 'ADMIN', '17:06:41', and '28-05-2022', and a 'HOME' button. On the left, a table lists users: ADMIN, OPERATOR, and b. Below the table are 'Add', 'Edit', and 'Delete' buttons. The main area contains fields for 'USER NAME', 'PASSWORD', and 'OLD PASSWORD', along with checkboxes for 'MANUAL OPERATION', 'MODEL SETTINGS', 'USER SETTINGS', 'COMMUNICATION SETTINGS', and 'REPORTS'. At the bottom right are 'Update' and 'Cancel' buttons. Numbered callouts (1) to (5) point to the HOME button, the USER information area, the main settings area, the Update button, and the Cancel button respectively.

User Name	Model Settings	User Settings
ADMIN	1	1
OPERATOR	1	1
b	1	1

USER NAME : ADMIN

PASSWORD :

OLD PASSWORD :

☒ MANUAL OPERATION

☒ MODEL SETTINGS

☒ USER SETTINGS

☒ COMMUNICATION SETTINGS

☒ REPORTS

Update Cancel

1.Home screen navigation button.

- User name, date and time display.

2.User name list display.

- Add, delete and edit navigation button.

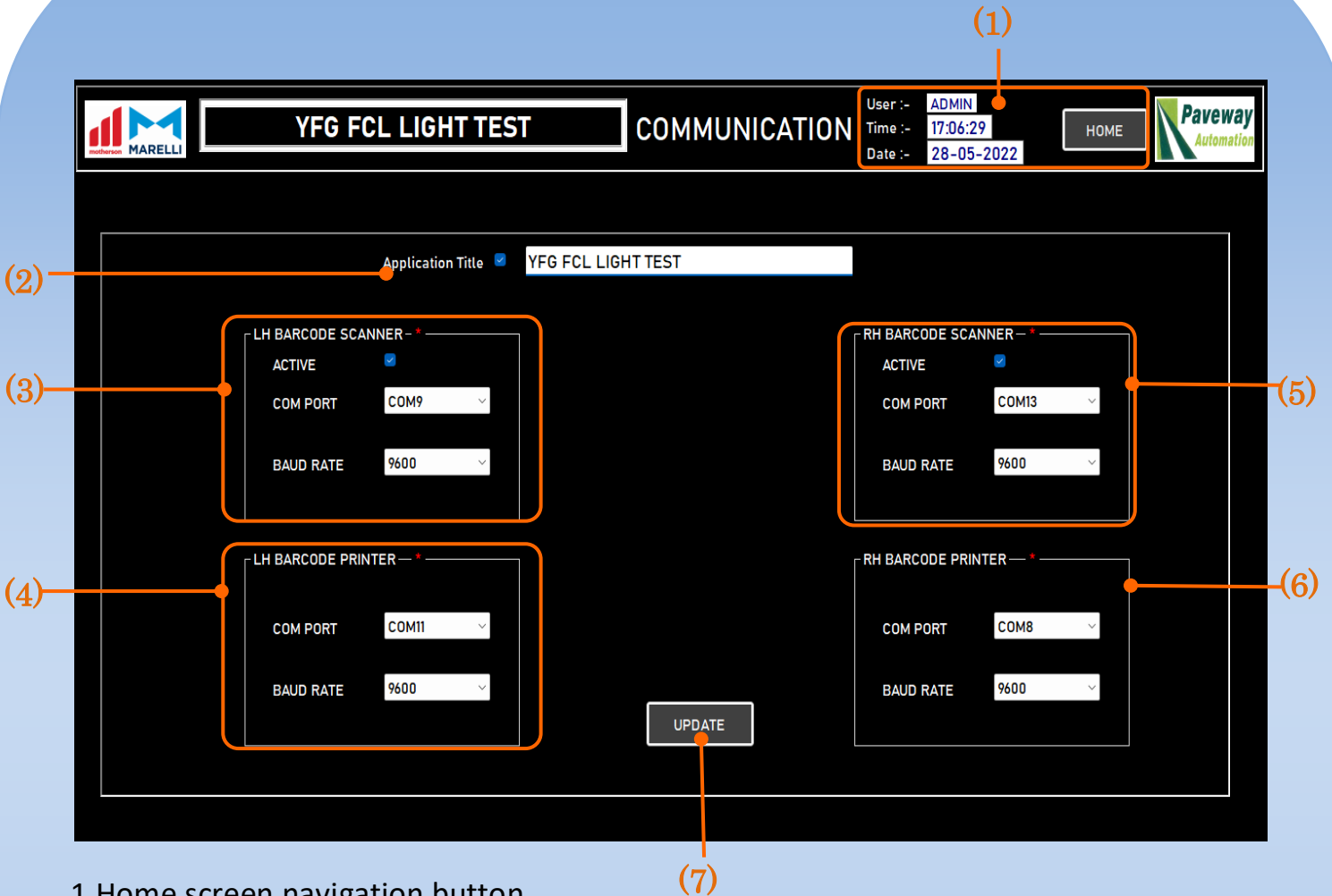
3. Enter username, password and old password display.

- Select manual operation, model settings, user settings, communication settings and reports.

4. Update navigation button.

5. Cancel navigation button.

COMMUNICATION SETTING SCREEN



The screenshot shows the 'COMMUNICATION SETTING SCREEN' for the 'YFG FCL LIGHT TEST' application. The interface includes a header with the application title and a 'HOME' button. Below the header, there are four configuration sections: 'LH BARCODE SCANNER', 'RH BARCODE SCANNER', 'LH BARCODE PRINTER', and 'RH BARCODE PRINTER'. Each section contains an 'ACTIVE' checkbox, a 'COM PORT' dropdown menu, and a 'BAUD RATE' dropdown menu. An 'UPDATE' button is located at the bottom center of the screen. Numbered callouts (1) through (7) point to the following elements:

- (1) User name, date, and time display.
- (2) Project title display and select/unselect option setting.
- (3) LH barcode scanner-com port and baud rate select list setting.
- (4) LH barcode printer-com port and baud rate select list setting.
- (5) RH barcode scanner-com port and baud rate select list setting.
- (6) RH barcode printer-com port and baud rate select list setting.
- (7) Update navigation button.

1.Home screen navigation button.

- User name, date and time display.

2.Project title display and select/unselect option setting.

3. LH barcode scanner-com port and baud rate select list setting.

4. LH barcode printer-com port and baud rate select list setting.

5. RH barcode scanner-com port and baud rate select list setting.

6. RH barcode printer- com port and baud rate select list setting.

7. Update navigation button.4. Update navigation button.



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REPORT SCREEN-1

YFG FCL LIGHT TEST

REPORTS

User :- **ADMIN**
Time :- **17:07:10**
Date :- **28-05-2022**

HOME

☒ All Models
☐ By Model
☐ By Primary_Barcode

FORM : **28-04-2022**
TO : **28-05-2022**

Export To Excel
Generate

Status

id	dte	Primary_Barcode	barcode	model	variant	leakrate	homolo	lamp lcurmin	lamp lcurmax	lamp lcuract	lamp lchrom
789	2022-05-04 14:20:29	35602-76TA0 LH 020522 B 0001 +00,65 HHOK	35602-76TA0LH04050222B0001	YFG FCL DER	LH	+00,65	HHOK	0.073	0.160	0.085	T1,0,1
790	2022-05-05 20:00:30	35602-76TA0 LH 050522 B 0002 +00,95 HHOK	35602-76TA0LH05050222B0002	YFG FCL DER	LH	+00,95	HHOK	0.073	0.160	0.088	T1,0,103
791	2022-05-05 20:25:38	35601-76TA0 RH 050522 B 0002 +00,98 HHOK	35601-76TA0RH05050222B0002	YFG FCL DER	RH	+00,98	HHOK	0.073	0.160	0.084	T1,1,003
792	2022-05-05 21:17:38	35601-76TA0 RH 020522 B 0001 +00,35 HHOK	35601-76TA0RH05050222B0001	YFG FCL DER	RH	+00,35	HHOK	0.073	0.160	0.087	T1,1,003
793	2022-05-05 21:45:05	35601-76TA0 RH 190422 B 0002 +00,45 HHOK	35601-76TA0RH05050222B0002	YFG FCL DER	RH	+00,45	HHOK	0.073	0.160	0.079	T1,1,003
794	2022-05-05 21:47:00	35601-76TA0 RH 050522 C 0002 +00,51 HHOK	35601-76TA0RH05050222B0002	YFG FCL DER	RH	+00,51	HHOK	0.073	0.160	0.080	T1,1,003
795	2022-05-05 21:51:41	35601-76TA0 RH 060522 C 0003 +00,50 HHOK	35601-76TA0RH05050222B0003	YFG FCL DER	RH	+00,50	HHOK	0.073	0.160	0.084	T1,1,003
796	2022-05-05 21:55:46	35601-76TA0 RH 060522 C 0004 +01,94 HHOK	35601-76TA0RH05050222B0004	YFG FCL DER	RH	+01,94	HHOK	0.073	0.160	0.083	T1,1,003
797	2022-05-05 22:18:53	35602-76TA0 LH 060522 C 0003 +00,87 HHOK	35602-76TA0LH05050222A0003	YFG FCL DER	LH	+00,87	HHOK	0.073	0.160	0.088	T1,0,1
798	2022-05-05 22:24:16	35601-76TA0 RH 060522 C 0006 +00,55 HHOK	35601-76TA0RH05050222A0006	YFG FCL DER	RH	+00,55	HHOK	0.073	0.160	0.083	T1,1,0
799	2022-05-05 22:30:07	35601-76TA0 RH 060522 C 0009 +00,44 HHOK	35601-76TA0RH05050222A0009	YFG FCL DER	RH	+00,44	HHOK	0.073	0.160	0.086	T1,1,0
800	2022-05-05 22:32:45	35602-76TA0 LH 060522 C 0009 +00,79 HHOK	35602-76TA0LH05050222A0009	YFG FCL DER	LH	+00,79	HHOK	0.073	0.160	0.087	T1,0,1
801	2022-05-05 22:38:03	35601-76TA0 RH 050522 B 0001 +00,73 HHOK	35601-76TA0RH05050222A0001	YFG FCL DER	RH	+00,73	HHOK	0.073	0.160	0.086	T1,1,0
802	2022-05-05 22:40:50	35602-76TA0 LH 050522 B 0001 +00,73 HHOK	35602-76TA0LH05050222A0001	YFG FCL DER	LH	+00,73	HHOK	0.073	0.160	0.086	T1,0,1
803	2022-05-06 09:39:04	35602-76TA0 LH 060522 A 0001 +01,45 HHOK	35602-76TA0LH06050222A0001	YFG FCL DER	LH	+01,45	HHOK	0.073	0.160	0.089	J,0,1
804	2022-05-06 09:51:11	35601-76TA0 RH 060522 C 0005 +01,92 HHOK	35601-76TA0RH06050222A0005	YFG FCL DER	RH	+01,92	HHOK	0.073	0.160	0.089	T1,1,0
805	2022-05-06 10:00:49	35601-76TA0 RH 060522 C 0008 +00,70 HHOK	35601-76TA0RH06050222A0008	YFG FCL DER	RH	+00,70	HHOK	0.073	0.160	0.085	T1,1,0
806	2022-05-06 10:03:35	35601-76TA0 RH 060522 C 0010 +01,71 HHOK	35601-76TA0RH06050222A0010	YFG FCL DER	RH	+01,71	HHOK	0.073	0.160	0.090	T1,1,0
807	2022-05-06 10:05:08	35601-76TA0 RH 060522 C 0011 +00,49 HHOK	35601-76TA0RH06050222A0011	YFG FCL DER	RH	+00,49	HHOK	0.073	0.160	0.086	T1,1,0
808	2022-05-06 11:32:50	35602-76TA0 LH 050522 B 0004 +01,15 HHOK	35602-76TA0LH06050222A0004	YFG FCL DER	LH	+01,15	HHOK	0.073	0.160	0.087	T1,0,1
809	2022-05-06 12:05:13	35602-76TA0 LH 060522 A 0003 +01,52 HHOK	35602-76TA0LH06050222A0003	YFG FCL DER	LH	+01,52	HHOK	0.073	0.160	0.084	T1,0,1

1.Home screen navigation button.

- User name, date and time display.

2.Select all models option.

- From to To select date.
- Export to excel option button.
- Generate option button.

3. Barcode data:-

- Barcode id, date, model, variant display.
- Leak rate values display,Homologation.
- Lamp 1 current minimum, maximum, actual parameter display.



YFG FCL LIGHT TEST.

Marelli Motherson Automotive Lighting India Pvt Ltd.
Gat No-165/1,165/2 &165/3 Village Ambethan,
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REPORT SCREEN-2

(1)

YFG FCL LIGHT TEST **REPORTS**

User :- ADMIN
Time :- 17:08:22
Date :- 28-05-2022

HOME

Export To Excel

Generate

(2)

All Models
By Model YFG FCL STD
By Primary_Barcode

FORM : 01-05-2022
TO : 11-05-2022

Status

(3)

id	dte	Primary_Barcode	barcode	model	variant	leakrate	homolo	lamp1curmin	lamp1curmax	lamp1curact	lamp1chrom
819	2022-05-06 18:21:47	35602-76T00 LH 060522 B 0001 +00,71 HHOK	35602-76T00LH06052022B0001	YFG FCL STD	LH	+00,71	HHOK	0.295	0.595	0.453	T1,0,0
820	2022-05-06 18:22:15	35601-76T00 RH 060522 B 0001 +00,67 HHOK	35601-76T00RH06052022B0001	YFG FCL STD	RH	+00,67	HHOK	0.295	0.595	0.442	T1,0,0
821	2022-05-06 19:41:54	35602-76T00 LH 060522 B 0003 +00,61 HHOK	35602-76T00LH06052022B0003	YFG FCL STD	LH	+00,61	HHOK	0.295	0.595	0.450	T1,0,1
822	2022-05-06 19:48:23	35602-76T00 LH 060522 B 0004 +00,79 HHOK	35602-76T00LH06052022B0004	YFG FCL STD	LH	+00,79	HHOK	0.295	0.595	0.449	T1,0,1
823	2022-05-06 19:51:58	35602-76T00 LH 060522 B 0005 +00,88 HHOK	35602-76T00LH06052022B0005	YFG FCL STD	LH	+00,88	HHOK	0.295	0.595	0.458	T1,0,1
824	2022-05-06 19:57:38	35601-76T00 RH 060522 B 0002 +01,17 HHOK	35601-76T00RH06052022B0002	YFG FCL STD	RH	+01,17	HHOK	0.295	0.595	0.443	T1,1,0
825	2022-05-06 20:00:22	35601-76T00 RH 060522 B 0005 +00,68 HHOK	35601-76T00RH06052022B0005	YFG FCL STD	RH	+00,68	HHOK	0.295	0.595	0.435	T1,1,0
826	2022-05-06 20:05:09	35601-76T00 RH 060522 B 0006 +02,61 HHOK	35601-76T00RH06052022B0006	YFG FCL STD	RH	+02,61	HHOK	0.295	0.595	0.444	T1,1,0
827	2022-05-06 20:06:56	35602-76T00 LH 060522 B 0006 +02,05 HHOK	35602-76T00LH06052022B0006	YFG FCL STD	LH	+02,05	HHOK	0.295	0.595	0.439	T1,0,1
828	2022-05-06 20:08:54	35602-76T00 LH 060522 B 0007 +01,64 HHOK	35602-76T00LH06052022B0007	YFG FCL STD	LH	+01,64	HHOK	0.295	0.595	0.451	T1,0,1
829	2022-05-06 20:10:59	35602-76T00 LH 060522 B 0008 +00,89 HHOK	35602-76T00LH06052022B0008	YFG FCL STD	LH	+00,89	HHOK	0.295	0.595	0.440	T1,0,1
830	2022-05-06 20:12:58	35602-76T00 LH 060522 B 0009 +00,65 HHOK	35602-76T00LH06052022B0009	YFG FCL STD	LH	+00,65	HHOK	0.295	0.595	0.439	T1,0,1
831	2022-05-06 20:16:39	35602-76T00 LH 060522 B 0010 +00,68 HHOK	35602-76T00LH06052022B0010	YFG FCL STD	LH	+00,68	HHOK	0.295	0.595	0.432	T1,0,1
832	2022-05-06 20:20:03	35602-76T00 LH 060522 B 0011 +01,35 HHOK	35602-76T00LH06052022B0011	YFG FCL STD	LH	+01,35	HHOK	0.295	0.595	0.441	T1,0,1
833	2022-05-06 20:23:33	35601-76T00 RH 060522 B 0007 +00,60 HHOK	35601-76T00RH06052022B0007	YFG FCL STD	RH	+00,60	HHOK	0.295	0.595	0.442	T1,1,0
834	2022-05-06 20:28:19	35602-76T00 LH 060522 B 0012 +00,80 HHOK	35602-76T00LH06052022B0012	YFG FCL STD	LH	+00,80	HHOK	0.295	0.595	0.441	T1,0,1
835	2022-05-06 20:34:36	35601-76T00 RH 060522 B 0008 +00,79 HHOK	35601-76T00RH06052022B0008	YFG FCL STD	RH	+00,79	HHOK	0.295	0.595	0.442	T1,1,0
836	2022-05-06 20:38:26	35602-76T00 LH 060522 B 0013 +00,74 HHOK	35602-76T00LH06052022B0013	YFG FCL STD	LH	+00,74	HHOK	0.295	0.595	0.443	T1,0,1
837	2022-05-06 20:41:06	35601-76T00 RH 060522 B 0004 +01,14 HHOK	35601-76T00RH06052022B0004	YFG FCL STD	RH	+01,14	HHOK	0.295	0.595	0.450	T1,1,0
838	2022-05-06 21:30:42	35601-76T00 RH 060522 C 0003 +00,71 HHOK	35601-76T00RH06052022B0003	YFG FCL STD	RH	+00,71	HHOK	0.295	0.595	0.445	T1,1,0
839	2022-05-06 21:35:05	35601-76T00 RH 070522 C 0004 +02,19 HHOK	35601-76T00RH06052022B0004	YFG FCL STD	RH	+02,19	HHOK	0.295	0.595	0.442	T1,1,0

1.Home screen navigation button.

- User name, date and time display.

2.Select by model option.

- From to To select date.
- Export to excel option button.
- Generate option button.

3. Barcode data:-

- Barcode id, date, model, variant display.
- Leak rate values display, Homologation.
- Lamp 1 current minimum, maximum, actual parameter display.



YFG FCL LIGHT TEST.

Marelli Motherson Automotive Lighting India Pvt Ltd.
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REPORT SCREEN-3

YFG FCL LIGHT TEST REPORTS

User :- ADMIN
Time :- 17:09:19
Date :- 28-05-2022

HOME

Export To Excel

Generate

☐ All Models
☐ By Model
☒ By Primary_Barcode

35602-76T00 LH 010422 B 0014 +00,57 HHOK

Status											
id	dte	Primary_Barcode	barcode	model	variant	leakrate	homolo	lamp1curmin	lamp1curmax	lamp1curact	lamp1chrom
875	2022-05-17 14:47:31	35602-76T00 LH 010422 B 0014 +00,57 HHOK	35602-76T00 LH 17052022 B 0014	YFG FCL STD	LH	+00,57	HHOK	0.295	0.595	0.450	T1.0.0

1.Home screen navigation button.

- User name, date and time display.

2.Select by primary barcode option.

- Export to excel option button.
- Generate option button.

3. Barcode data:-

- Barcode id, date, model, variant display.
- Leak rate values display,
- Homologation.
- Lamp 1 current minimum, maximum, actual parameter display.



YFG FCL LIGHT TEST.

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MACHINE ERRORS AND REMEDY

SR. NO	ERROR DESCRIPTION	SR. NO	CAUSES OF ERROR	SR. NO	REMEDY
1	EMERGENCY PB PRESSED	1	Emergency push button might have pressed.	1	Release emergency push button and press Reset push button to clear the error.
		2	There may be loose connection in the push button contact.	2	Tighten the electrical connection and press Reset push button to clear the error.
2	AIR PRESSURE NOT IN RANGE	1	Check the pneumatic supply.	1	Adjust the pressure to optimum level.
		2	There may be loose connection in the pressure indicator.	2	Tighten the connector and press Reset push button to clear the error.
3	FIXTURE ERROR	1	Check loaded fixture and set fixture in hmi are same or different.	1	Go in configuration screen then click on fixture number button to set actual fixture number.
		2	Check input indication on plc there may be loose connection in the electrical connector.	2	Tighten the electrical connection and press Reset push button to clear the error.
4	LIGHT CURTAIN INTRRUPTED	1	There may be any interruption in light curtain	1	Check any interruption in between light curtain receiver and transmitter
		2	There may be chance of disturb light curtain mounting arrangement		Please check is transmitter and receiver is in align
5	PLACE NEW PART	1	After completing process new process is not taken with new part	1	Please take new process with new cycle
		2	The sensor position might have shifted.	2	Position the sensor properly.
6	PART 1,2,3,4,5,6 NOT PRESENT	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
7	CYL FWD ERROR	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error



YFG FCL LIGHT TEST.

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		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
8	CYL RET ERROR	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
9	LAMP VOLTAGE CHECK ERROR	1	Lamp actual voltage and lamp set voltage may be different.	1	Go in configuration screen, check set voltage are same or different as compare to actual voltage.
		2	There may be loose connection in the electrical meeting connector.	2	Tighten the electrical connection properly and press reset push button to clear the error
		3	There may be any problem in DC power supply.	3	Check voltage on DC power supply via through external multimeter.
		4	There may be any problem in voltmeter.	4	Check voltage on voltmeter.
10	LAMP CURRENT CHECK ERROR	1	Lamp actual current and lamp set current may be different.	1	Go in configuration screen, check set current are same or different as compare to actual current.
		2	There may be loose connection in the electrical meeting connector.	2	Tighten the electrical connection properly and press reset push button to clear the error
		3	There may be any problem in ammeter.	4	Check current on ammeter.
11	LAMP IMAGE ERROR	1	Image captured by camera is not match with master image	1	check part placement & fixture alignment properly
		2	There may be chance of part is not ok	2	Please check part with master part
		3	lighting arrangement of camera may be disturbed	4	Please check lighting arrangement of camera
6	BARCODE NOK	1	The scan data is not match with actual set data in configuration screen	1	Check scan barcode data is ok or not
		2	The scanner may not be able to find the right data after scanning.	2	Check connection of barcode scanner or barcode scan properly.

YFG FCL LIGHT TEST.



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		3	There may be loose communication cable connection.	3	Check connection of scanner are connect correct or wrong.
		4	There may be barcode are not scan properly.	4	Check the proper condition of the barcode and scan properly.
16	CAMERA COMMUNICATION ERROR	1	There may be loose connection in communication cable.	1	Check communication cable between camera controller and plc.
17	CAMERA NOT IN RUN MODE	1	Camera is in setup mode	1	Please press camera run mode tab in camera application
18	POST SCAN BARCODE ERROR	1	The scan data is not match with actual set data in configuration screen	1	Check scan barcode data is ok or not
		2	The scanner may not be able to find the right data after scanning.	2	Check connection of barcode scanner or barcode scan properly.
		3	There may be loose communication cable connection.	3	Check connection of scanner are connect correct or wrong.
19	SERVER NOT CONNECTED	1	Server and plc communication is lost	1	check communication cable between plc and server
		2	There may be loose communication cable connection.	2	check communication cable between plc and server
		3	server application not open	4	Open server application and establish connection again.
20	BARCODE NOT AVIALABLE IN DATABASE	1	May be previous station process is not completed	1	Please complete previous station process
		2	Barcode already logged in database and final barcode generated.	2	final barcode is generated
21	SERVER BUSY	1	Server is already busy in communication	1	Please wait for communication result and retry server connection.

YFG FCL LIGHT TEST.



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DO'S & DONT'S:

Every effort has been taken to ensure user safety while designing this machine. However, you are requested to adhere to following instructions to avoid any mishap.

- The machine should not be used for any other purpose other than that specified in the intended use.
- Do not operate the machine without understanding the installation, operation and maintenance manual thoroughly.
- Only trained, skilled, qualified and authorized personnel should install, operate and maintain the machine.
- Do not commission the machine without the aid of the qualified instructor.
- Do not reach into moving machinery.
- Never perform complex maintenance procedure single handed. Especially during start-ups as it is necessary to have another person manning the emergency stop in case of any problem.
- No tools other than that are prescribed should be used for machine maintenance.
- Do not store any loose tools in the machine compartments or control panel.
- If the machine is down for maintenance, place a sign over the control panel to inform others about that.
- Switch off the machine while connecting/disconnecting into electrical connector.

YFG FCL LIGHT TEST.



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